数理分析方法

连续对应与最优化

Lecture 108

原课件逐页保留 · 零基础中文讲解 · 证明训练

材料与编写说明

课程原页：张顺明教授课件（用户提供）。参考：Gérard Debreu, Theory of Value (1959)，第一章 Mathematics。原页与本讲义的补充、纠错、例题分开呈现。本文为辅助自学材料，不代表教师官方讲义。

怎样阅读

先看本页的中文说明，再对照原页符号；遇到证明，先抄出假设和目标，再逐行说明“这一步用的是什么”。遮住证明重写一遍，最后用反例检查条件是否必要。重复或动画展开页仍逐页保留；说明会指出新增步骤或对应的完整论证位置。

“补充”用于补基础或展开原课件略去的论证；“纠错”指出原页需要修正的地方。课程概览及人物介绍页按其原有用途解释，不虚构数学定理。

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零基础读证明：随手查阅

对象的层次。 $x \in A$ 表示一个元素属于集合，$A \subseteq B$ 表示 A 的每个元素都在 B 中。$\{x\}$ 是只含 x 的集合，不是 x 本身。$\varnothing$ 是空集，$\{\varnothing\}$ 则含有一个元素。$\mathbb{R}^N$ 中的点是有 N 个坐标的有序列表。

量词。 $\forall$ 是“对所有”，$\exists$ 是“至少存在一个”。例如“每个学生有一本书”不要求是同一本；“有一本书供每个学生使用”则要求同一本。证明极限时 $\forall \varepsilon > 0\ \exists n_0\ \forall n \geq n_0$ 的顺序不可换：先指定精度，再找统一起点，最后控制这个起点之后的全部项。

怎样否定。 否定“所有 x 满足 P”得到“存在 x 不满足 P”；否定“存在”得到“所有都不”。否定 $P \Rightarrow Q$ 是 P 真且 Q 假。因此要推翻一个定理，必须满足其全部假设又违反结论；拿不满足假设的例子不能当反例。

四种基本证明。 (1) 包含 $A \subseteq B$：任取 $x \in A$，用定义得到 $x \in B$。(2) 相等 $A = B$：两个方向分别证明。(3) 等价 $P \iff Q$：证明 $P \Rightarrow Q$ 及反向。(4) 反证：把结论否定加入已知，推出矛盾。逆否命题 $\neg Q \Rightarrow \neg P$ 与 $P \Rightarrow Q$ 等价，“反过来” $Q \Rightarrow P$ 则未必成立。

数列、子列和收敛。 数列 x_n 是第 n 个对象；子列 x_{n_k} 的指标必须严格递增。$x_n \to x$ 指任意小的误差最终都能保持。前面有限多项不影响极限。抽子列通常是为了用有界性和有限维 Bolzano–Weierstrass 定理获得极限；抽取之后须把原来所有相关量一起限制到同一子列。

距离的关键估计。 $\|x\| = (\sum_i x_i^2)^{1/2}$，$\|x - y\|$ 是距离。三角不等式 $\|x - z\| \leq \|x - y\| + \|y - z\|$ 允许经中间点拆误差。两段误差各小于 $\varepsilon/2$，总误差便小于 ε；分成 m 段时可各分配 ε/m。这里的 ε 不是固定的小数，而是任意正精度。

做证明题的五行草稿。 先写完整假设；写准确目标；列出可用定义与定理；逐步注明所用条件；最后检查任意性、存在性和量词顺序。出现“显然”时问：若阅卷者要求展开，我能否补出一个集合等价链、一个不等式或一个明确构造？不知道证明的命题要标为引用，不能把待证结论循环当作理由。

本讲先修与使用方法

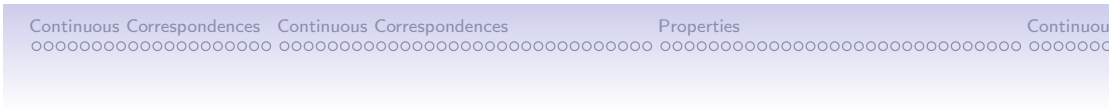
本讲把普通函数推广为“一个输入对应一整套允许结果”的**对应（correspondence）**。例如预算变化时，可买的商品组合组成一个集合。全文用 $\varphi : S \rightrightarrows T$ 表示对应，$\varphi(x) \subseteq T$ 是它在 x 处的值。除特别讨论空集的页外，默认值非空。$x_n \to x$ 表示输入越来越接近 x；$y_n \in \varphi(x_n)$ 表示每次都选择一个当时允许的结果。上标 q 在原课件里通常只是序列编号，不是乘方。

四个基础词。 开集的每个点周围都有一个小球仍在集合中；闭集包含其所有收敛序列的极限；在有限维欧氏空间，紧集等价于闭且有界，也等价于每个序列都有收敛子列且极限仍在集合中；邻域是包含该点某个开球的集合，本文可直接取开邻域。$\varphi(A) = \bigcup_{x \in A} \varphi(x)$ 是所有输入在 A 中时能出现的结果。

先分清三种性质。 闭图：输入和结果已经一起收敛时，极限仍然可行。上半连续（开集定义）：包含当前全部结果的开集，也会包含附近的全部结果。下半连续：当前的每个结果，附近都能找到接近它的可行结果。闭图与上半连续并非一般等价；第4页的共同紧值域条件使它们等价。课件前后切换定义，读题时必须先写清假设。第84页把一个“必能提取子列”的强性质称为 o.s.c.，本讲按课件使用这一名称并明确提醒它与常见术语的差异。

建议学习顺序。 先学原页4–14的定义与序列证明，再学15–21的最大值定理；23–53补足开集语言及紧性；54–95训练运算与纠错；96–113系统整理最优化。原页每页均完整保留；下方文字是针对该页的中文说明。课件中的错误论证不能作为考试答案，标有“纠正”的地方给出可用版本。参考背景为 Debreu, Theory of Value 第1章第8节；后半部分的推广与纠错是学习补充，不冒充教材原话。

1 原课件第 1 页：从连续函数走向连续对应



Theory of Value

An Axiomatic Analysis of Economic Equilibrium
Gerard Debreu (1959)

Professor Shunming Zhang

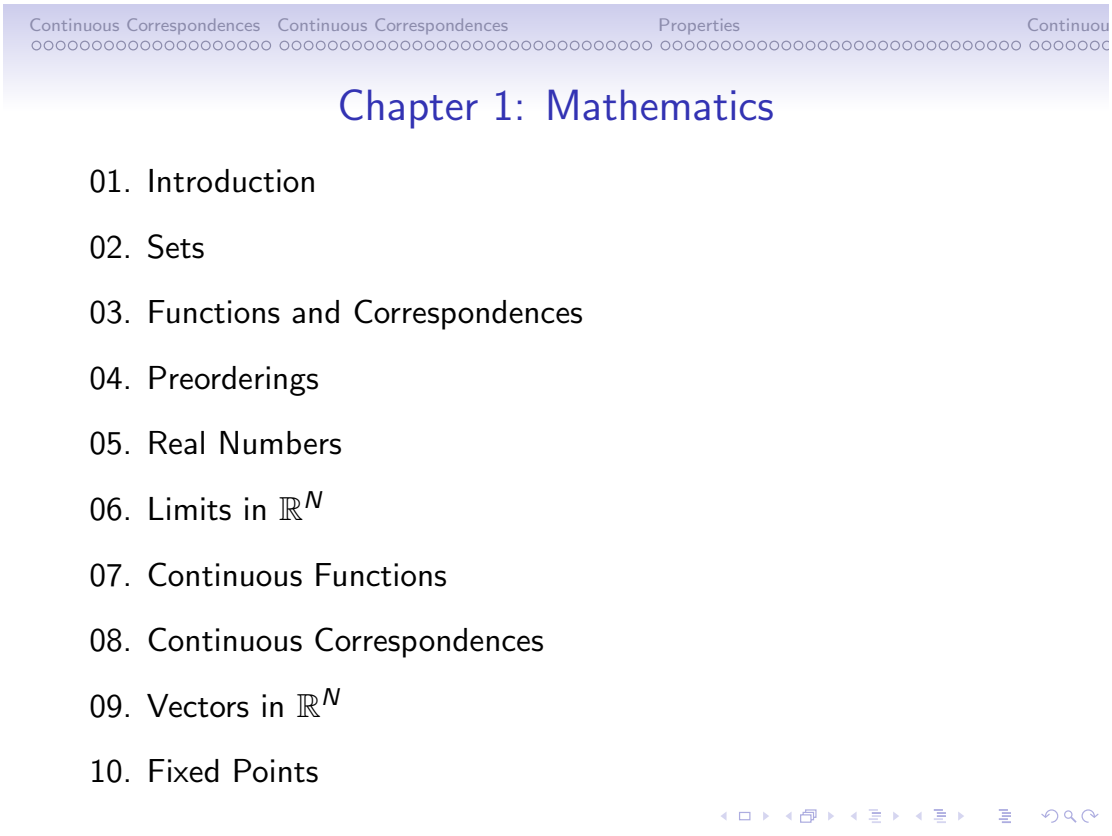
School of Finance
Renmin University of China

October 1, 2022

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

标题页确定主题：连续对应。普通函数 $f(x)$ 是一个点，对应 $\varphi(x)$ 是一个集合。若一名消费者有多个并列最优选择，用函数会丢失信息，对应能把这些选择全部保留下来。本讲的主问题是：环境轻微变化时，可行集合、最高收益、最优选择分别怎样变化?

2 原课件第 2 页：本讲在全章中的位置



上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

集合与映射提供语言，实数与极限提供收敛，连续函数提供“带入极限”的规则；本讲把这些规则用于集合值映射。下一讲的凸性和最后的固定点定理将使用本讲的上半连续对应。现阶段务必熟悉 $\in$ 与 $\subseteq$：$y \in \varphi(x)$ 是点属于集合，而 $\varphi(x) \subseteq G$ 是整套结果都在 G 中。

3 原课件第 3 页：经济学为什么需要对应

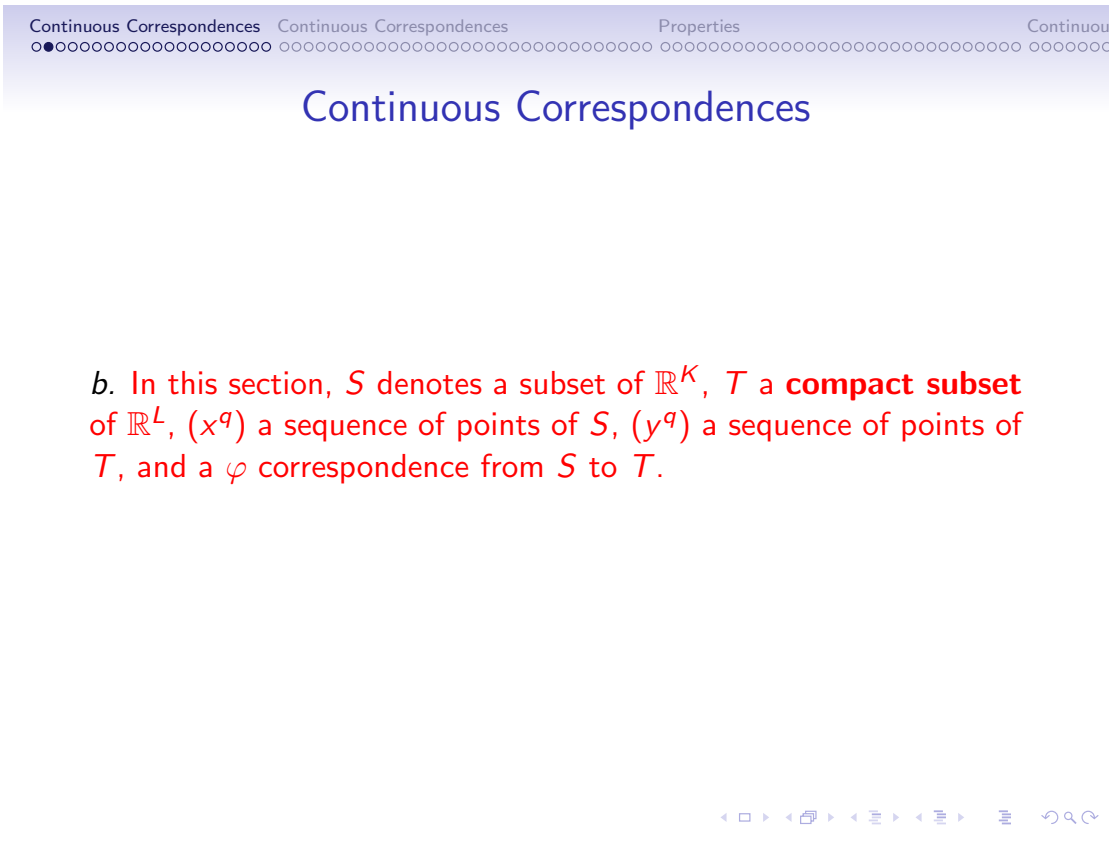


a. Example (2) of 1.3.h has pointed out the interest of correspondences for economics; this section will study their continuity.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

环境 x 可以包含价格、收入和技术；$\varphi(x)$ 是这一环境下允许的行动集合。例如 $\varphi(p,w)=\{y\geq 0: p\cdot y\leq w\}$ 是预算集。研究连续性不是要求集合内每个点保持不变，而是要求变化遵守某种“可逼近”的规律。非空性、紧性与连续性在优化中承担不同职责：有选择、能取到最优、最优行为对参数稳定。

4 原课件第 4 页：本节共同假设：值域紧



上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

课件先取 $S \subseteq \mathbb{R}^K$、**紧集** $T \subseteq \mathbb{R}^L$，且 $\varphi(x) \subseteq T$。这意味着所有可能结果都装在同一个闭有界容器里。若 $y_n \in \varphi(x_n)$，不管 x_n 如何变化，总可从 $y_n \in T$ 抽出收敛子列。这一点比“每个 $\varphi(x)$ 单独紧”强：$\varphi(x) = \{1/x\}$（$x > 0$）每个值都紧，但 $x \downarrow 0$ 时结果会逃到无穷远。后续不用共同紧 T 时，需要真正的上半连续加紧值性补上控制。

5 原课件第 5 页：连续性拆成两个方向



c. The concept of continuity for a correspondence will be introduced in three steps.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

上半连续控制“不能在附近出现脱离当前集合的结果”；下半连续控制“当前结果不能在附近突然无法逼近”。两者合起来才是连续对应。不要把“上、下”理解为图像往上或往下，它们约束的是整套集合变化的不同方向。最优解对应往往只有上半连续，最优值却仍可连续，这正是本讲的核心现象。

6 原课件第 6 页：课件的上半连续：闭图式序列条件

Continuous Correspondences

d. Let x^0 be a point of S; one defines firstly:
The correspondence φ is upper semicontinuous (u.s.c.) at the point x^0 if:

"$x^q \to x^0, y^q \in \varphi\left(x^q\right), y^q \to y^0$" implies "$y^0 \in \varphi\left(x^0\right)$."

In other words, if x^q tends to x^0, and if y^q tends to y^0 while belonging for all q to the image-set of x^q, one requires that y^0 belong to the image-set of x^0. One could also say, if $x^q \to x^0$, and if the distance from y^0 to the image-set of x^q tends to zero, one requires that y^0 belong to the image-set of x^0. This is a natural but rather weak continuity requirement, as the following example shows. Let S and T be two compact real intervals. The graph of φ (fig. 2.a) is the shaded region, heavy-lined boundary included; $\varphi\left(x^0\right)$ is the interval $\left[a^1, a^2\right]$. The correspondence φ is closed at x^0.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

课件条件逐字读作：对任意 $x_n \to x_0$，对任意 满足 $y_n \in \varphi(x_n)$ 且 $y_n \to y_0$ 的选择，都有 $y_0 \in \varphi(x_0)$。结果序列收敛是前提，不能自行假定。它就是闭图的序列条件；在第4页共同紧 T 和闭值背景下，等价于后面的开集式上半连续。

距离说法也可验证：若 $d(y_0, \varphi(x_n)) \to 0$，选 $y_n \in \varphi(x_n)$ 使

$$\|y_n - y_0\| < d(y_0, \varphi(x_n)) + 1/n.$$

右边趋于零，故闭图条件给 $y_0 \in \varphi(x_0)$。为什么加 $1/n$？距离是下确界，未必已取到最小值，近似选择总能做到。

7 原课件第 7 页：图2：放大与缩小的区别

Continuous Correspondences

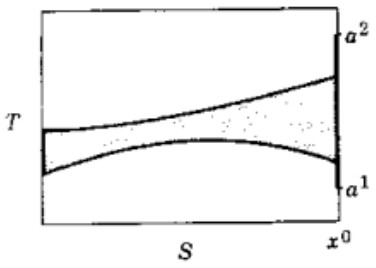


Fig. 2.a

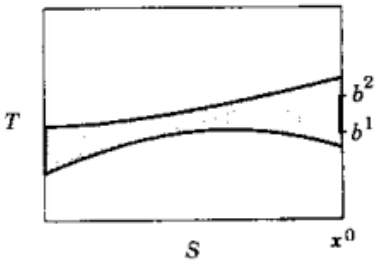


Fig. 2.b

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

图的横轴是输入，纵向截面才是 $\varphi(x)$，阴影表示图形 $G(\varphi)$，不要把它当普通函数曲线。左图在 x_0 的截面 $[a^1, a^2]$ 可以比邻近截面更大：附近的极限仍落在这一大截面内，因而可以上半连续，但大截面中新增的点未必能从附近逼近。右图在 x_0 只保留较小截面 $[b^1, b^2]$：其中每点能从附近找到近似点，可以下半连续，但邻近多出来的点的极限可能不在截面中。端点的粗细和是否包含是判断闭图的关键。

8 原课件第 8 页：下半连续：为指定目标构造逼近

Continuous Correspondences

e. One defines secondly:
The correspondence φ is lower semicontinuous (l.s.c.) at the point x^0 if:

"$x^q \to x^0, y^0 \in \varphi\left(x^0\right)$" implies "there is $\left(y^q\right)$ such that $y^q \in \varphi\left(x^q\right), y^q \to y^0$.

In other words, if x^q tends to x^0, and if y^0 belongs to the image-set of x^0, one requires that there be an infinite sequence $\left(y^q\right)$ such that y^q tends to y^0 while belonging for all q to the image-set of x^q. One could also say, if x^q tends to x^0, and if y^0 belongs to the image-set of x^0, one requires that the distance from y^0 to the image-set of x^q tend to zero. Again, this is a natural but weak continuity requirement as the example of fig. 2.b shows. Here $\varphi\left(x^0\right)$ is the interval $\left[b^1, b^2\right]$. The correspondence φ is lower semi-continuous at x_0. Notice how the roles of "$y^0 \in \varphi\left(x^0\right)$" and "$y^q \in \varphi\left(x^q\right)$, $y^q \to y^0$" are permuted in the two preceding definitions.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

这里量词是：任给 $x_n \to x_0$，再任给 $y_0 \in \varphi(x_0)$，存在 一列 $y_n \in \varphi(x_n)$ 使 $y_n \to y_0$。与第6页区别是，现在目标点先给定，我们负责选择合适的 y_n；不是说所有选择都趋向它。

例如常值对应 $\varphi(x) = [0,1]$ 下半连续，因为对任意 $y_0 \in [0,1]$ 直接取 $y_n = y_0$。下半连续与 $d(y_0, \varphi(x_n)) \to 0$ 等价：一方向用 $d \le \|y_n - y_0\|$；另一方向按第6页加 $1/n$ 近似取点。距离趋零不要求 y_0 本身在每个邻近集合内。

9 原课件第 9 页：连续对应的定义



f. Finally one defines:

The correspondence φ is continuous at the point x^0 if it is upper and lower semicontinuous at x^0.

φ then has all the desirable continuity properties at x^0.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

连续对应等于同时上半连续和下半连续。一个基本例子是 $\varphi(x) = [a(x), b(x)]$，其中 a, b 连续且 $a \le b$。若 $y_n \in [a(x_n), b(x_n)]$ 且 $y_n \to y$，取极限得 $a(x) \le y \le b(x)$；若指定 $y \in [a(x), b(x)]$，取 $y_n = \min\{b(x_n), \max\{a(x_n), y\}\}$，则 y_n 可行且趋于 y。两方向分别验证，才算证明完连续。

10 原课件第 10 页：单值对应退化为普通连续函数



g. When, for all $x \in S$, $\varphi(x)$ consists of a single element, the definition of lower semicontinuity at x^0 is obviously equivalent to the definition of continuity at x^0 for a function. One can prove that, in the same case, the definition of upper semicontinuity at x^0 is equivalent to the definition of continuity at x^0 for a function.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

设 $\varphi(x) = \{f(x)\}$。下半连续中的可选点只有 $y_n = f(x_n)$，故下半连续恰好要求 $f(x_n) \to f(x_0)$。对课件的闭图式上半连续，还要用紧 T：若 $f(x_n)$ 不趋于 $f(x_0)$，存在 $\varepsilon > 0$ 和子列始终距离至少 ε；紧性再抽出 $f(x_{n_k}) \to y$，闭图迫使 $y = f(x_0)$，矛盾。

去掉共同紧值域，闭图不够：$f(0) = 0$、$f(x) = 1/x$（$x \neq 0$）图闭，但在0不连续。若直接采用第24页开集定义，单值上半连续与普通连续的等价不需紧性。

11 原课件第 11 页：上半连续与闭图：双向证明

Continuous Correspondences

h. Semicontinuities and continuity on S are defined as semicontinuities and continuity at every point of S. It is clear that:

(1) The correspondence φ is upper semicontinuous on S if and only if its graph is closed in $S \times T$.

Proof. If the graph $G(\varphi)$ is closed in $S \times T$, then the correspondence φ is upper semicontinuous on S. For any $x^0 \in S$, if $x^q \to x^0$, $y^q \in \varphi(x^q)$, and $y^q \to y^0$, then $(x^q, y^q) \in G(\varphi)$ and $(x^q, y^q) \to (x^0, y^0)$. As $G(\varphi)$ is closed, then $(x^0, y^0) \in G(\varphi)$ and hence $y^0 \in \varphi(x^0)$.

Proof. If the correspondence φ is upper semicontinuous on S, then its graph is closed in $S \times T$. If $(x^q, y^q) \in G(\varphi)$ and $(x^q, y^q) \to (x^0, y^0)$, then $x^q \to x^0$, $y^q \to y^0$, and $y^q \in \varphi(x^q)$. As φ is upper semicontinuous on x^0, then $y^0 \in \varphi(x^0)$ and hence $(x^0, y^0) \in G(\varphi)$. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

图形定义为 $G(\varphi) = \{(x, y) : x \in S,\ y \in \varphi(x)\}$，闭是相对于 $S \times T$ 而言。若图闭，取 $x_n \to x_0$、$y_n \to y_0$ 且 $y_n \in \varphi(x_n)$；于是 $(x_n, y_n) \in G$ 且收敛至 (x_0, y_0)，闭性给 $(x_0, y_0) \in G$，翻译回去就是 $y_0 \in \varphi(x_0)$。

反之，取图中任意收敛序列 $(x_n, y_n) \to (x_0, y_0) \in S \times T$。乘积空间收敛等价于每个坐标收敛，课件序列条件给 $y_0 \in \varphi(x_0)$，所以极限仍在图中，图闭。**考试提醒：**若题目中的 u.s.c. 用开集定义，引用此结论时要同时说明紧值域或适当紧性假设。

12 原课件第 12 页：连续函数与对应的复合

Continuous Correspondences

i. As in 1.7.c let S_1, S_2, T be subsets of $\mathbb{R}^{K_1}$, $\mathbb{R}^{K_2}$, $\mathbb{R}^{L}$ respectively, f be a function from S_1 to S_2, and φ be a correspondence from S_2 to T. Define a correspondence ψ from S_1 to T by $\psi(x) = \varphi(f(x))$ for every x in S_1. It is immediate that:

(2) If f is continuous at the point x of S_1, and φ is upper semicontinuous (resp. lower semicontinuous) at the point $f(x)$ of S_2, then ψ is upper semicontinuous (resp. lower semicontinuous) at x.

Proof. If $x^q \to x$, $y^q \in \psi(x^q)$, and $y^q \to y$, as$f$ is continuous at the point x, then $z^q \equiv f(x^q) \to f(x) \equiv z$,
$y^q \in \psi(x^q) = \varphi(f(x^q)) \equiv \varphi(z^q)$. As φ is upper semicontinuous at the point $z \equiv f(x)$, then $y \in \varphi(z) = \varphi(f(x)) = \psi(x)$. Hence ψ is upper semicontinuous at x.
Proof. If $x^q \to x$, and $y \in \varphi(x)$, as$f$ is continuous at the point x, then $z^q \equiv f(x^q) \to f(x) \equiv z$, $y \in \psi(x) = \varphi(f(x)) \equiv \varphi(z)$. As φ is lower semicontinuous at the point $z \equiv f(x)$, then there is (y^q) such that $y^q \in \varphi(z^q) \equiv \varphi(f(x^q)) = \psi(x^q)$ and $y^q \to y$. Hence ψ is lower semicontinuous at x. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

定义 $\psi = \varphi \circ f$，即先由 x 得到点 $f(x)$，再得到集合 $\varphi(f(x))$。上半连续的证明是把中间变量写出：$x_n \to x$ 推出 $z_n = f(x_n) \to f(x) = z$；若 $y_n \in \psi(x_n) = \varphi(z_n)$ 且 $y_n \to y$，由 φ 的条件得 $y \in \varphi(z) = \psi(x)$。

下半连续：任取 $y \in \psi(x) = \varphi(z)$；由 $z_n \to z$，可选 $y_n \in \varphi(z_n)$ 且 $y_n \to y$，这些点正属于 $\psi(x_n)$。**纠正：**课件第二段开头的 $y \in \varphi(x)$ 应为 $y \in \psi(x)$；x 与 $f(x)$ 所在空间不同，不能混用。

13 原课件第 13 页：有限个对应的笛卡尔积

Continuous Correspondences

j. As in 1.7.d-e, for every $n = 1, 2, \cdots, N$ let T_n be a subset of $\mathbb{R}^{L_n}$, and φ_n be a correspondence from S to T_n. Consider the product $T = \prod_{n=1}^{N} T_n$ and define a correspondence φ from S to T by $\varphi(x) = \prod_{n=1}^{N} \varphi_n(x)$ for every x in S. If every T_n is compact, so is T by (7) of 1.6.t. In this case it is immediate that:

(3) If every φ_n is upper semicontinuous (resp. lower semicontinuous) at x of S, then φ is upper semicontinuous (resp. lower semicontinuous) at x.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$\varphi(x) = \prod_{i=1}^{N} \varphi_i(x)$ 的元素是向量 $(y_1, \ldots, y_N)$，每个坐标独立满足 $y_i \in \varphi_i(x)$。这常用于多个消费者同时选择：每人一个可行集，社会的可行选择组合是它们的积。有限积的连续性可逐坐标证明；第4页框架中，各个 T_i 紧则乘积也紧。有限维与有限个坐标使“各坐标收敛”等价于“整个向量收敛”。

14 原课件第 14 页：积对应的证明逐步写

Continuous Correspondences

Proof. If $x^q \to x$, $y^q \in \varphi(x^q)$, and $y^q \to y$, then $(y_1^q, \cdots, y_N^q) = y^q \in \varphi(x^q) = \prod_{n=1}^{N} \varphi_n(x^q)$ and $(y_1^q, \cdots, y_N^q) = y^q \to y = (y_1, \cdots, y_N)$, that is, $y_n^q \in \varphi_n(x^q)$ and $y_n^q \to y_n$ for $n = 1, \cdots, N$. As φ_n is upper semicontinuous at x, then $y_n \in \varphi_n(x)$ for $n = 1, \cdots, N$. Therefore $y = (y_1, \cdots, y_N) \in \prod_{n=1}^{N} \varphi_n(x) = \varphi(x)$. Hence φ is upper semicontinuous at x.

Proof. If $x^q \to x$, and $y \in \varphi(x)$, then $(y_1, \cdots, y_N) = y \in \varphi(x) = \prod_{n=1}^{N} \varphi_n(x)$, that is, $y_n \in \varphi_n(x)$ for $n = 1, \cdots, N$. As φ_n is lower semicontinuous at the point x, then there is (y_n^q) such that $y_n^q \in \varphi_n(z^q)$ and $y_n^q \to y_n$ for $n = 1, \cdots, N$. Thus $y^q = (y_1^q, \cdots, y_N^q) \in \prod_{n=1}^{N} \varphi_n(x^q) = \varphi(x^q)$ and $y^q = (y_1^q, \cdots, y_N^q) \to (y_1, \cdots, y_N) = y$. Hence φ is lower semicontinuous at x. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

上半连续：任取 $x_q \to x$、$y^q \in \varphi(x_q)$、$y^q \to y$。对每个 i，有 $y_i^q \in \varphi_i(x_q)$ 且 $y_i^q \to y_i$；上半连续得 $y_i \in \varphi_i(x)$，全部坐标同时成立即 $y \in \prod_i \varphi_i(x)$。

下半连续：给定 $y = (y_1, \ldots, y_N) \in \varphi(x)$。对每个 i，下半连续允许选 $y_i^q \in \varphi_i(x_q)$ 趋于 y_i；合成 $y^q = (y_1^q, \ldots, y_N^q)$，它可行，且 $\|y^q - y\|^2 = \sum_i |y_i^q - y_i|^2 \to 0$。**纠正：**课件的 $\varphi_n(z^q)$ 应为 $\varphi_n(x^q)$。

15 原课件第 15 页：可行集、最优值、最优解是三个对象

Continuous Correspondences

k. According to example (2) of 1.3.h. the interest of these concepts for economics lies, in particular, in the interpretations of an element x of S as the environment of a certain agent, of T as the set of actions a priori available to him, and of $\varphi(x)$ (assumed here to be closed for every x in S) as the subset of T to which his choice is actually restricted by his environment x. Let f be a continuous real-valued function on $S \times T$, and interpret $f(x, y)$ as the gain for that agent when his environment is x and his action y. Given x, one is interested in the elements of $\varphi(x)$ which maximize f (now a function of y alone) on $\varphi(x)$; they form a set $\mu(x)$. What can be said about the continuity of the correspondence μ from S to T?

$$\mu(x) = \{y \in \varphi(x) \mid y \text{ maximizes } f(x, y)\}$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

给定参数 x，可行行动是 $y \in \varphi(x)$，收益是连续函数 $f(x,y)$。定义

$$g(x) = \max_{y \in \varphi(x)} f(x,y), \qquad \mu(x) = \{y \in \varphi(x) : f(x,y) = g(x)\}.$$

$g(x)$ 是一个数，$\mu(x)$ 是一个集合。为保证它们有意义，必须假设 $\varphi(x)$ 非空且紧；本节共同紧 T 下，闭值即可得紧值。极值定理说明最大值确实取到，故 $\mu(x) \neq \varnothing$。只有上确界存在不够，例如在 $(0,1)$ 最大化 y 没有最优点。

16 原课件第 16 页：图3：最优解集嵌在可行集里

Continuous Correspondences

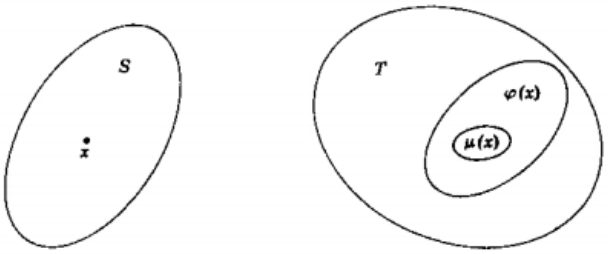


Fig. 3

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

图中左侧是环境空间 S 中的点 x，右侧依次是行动空间 T、可行集 $\varphi(x)$ 和最优解集 $\mu(x)$，表达

$$\mu(x) \subseteq \varphi(x) \subseteq T.$$

图中画成椭圆只是示意，不代表这些集合必为凸集、连通集或有内部。多个分开的点也可以同时最优；例如在 $[-1,1]$ 最大化 y^2，最优集是 $\{-1,1\}$。

17 原课件第 17 页：Berge最大值定理的结论

Continuous Correspondences

One is also interested in $g(x)$, the value of the maximum of f on $\varphi(x)$ for a given x.

$$g(x) = \max\{f(x,y) \mid y \in \varphi(x)\}$$

What can be said about the continuity of the real-valued function g on S? An answer to these two questions is given by the following result (the proof of continuity of g should not be attempted).

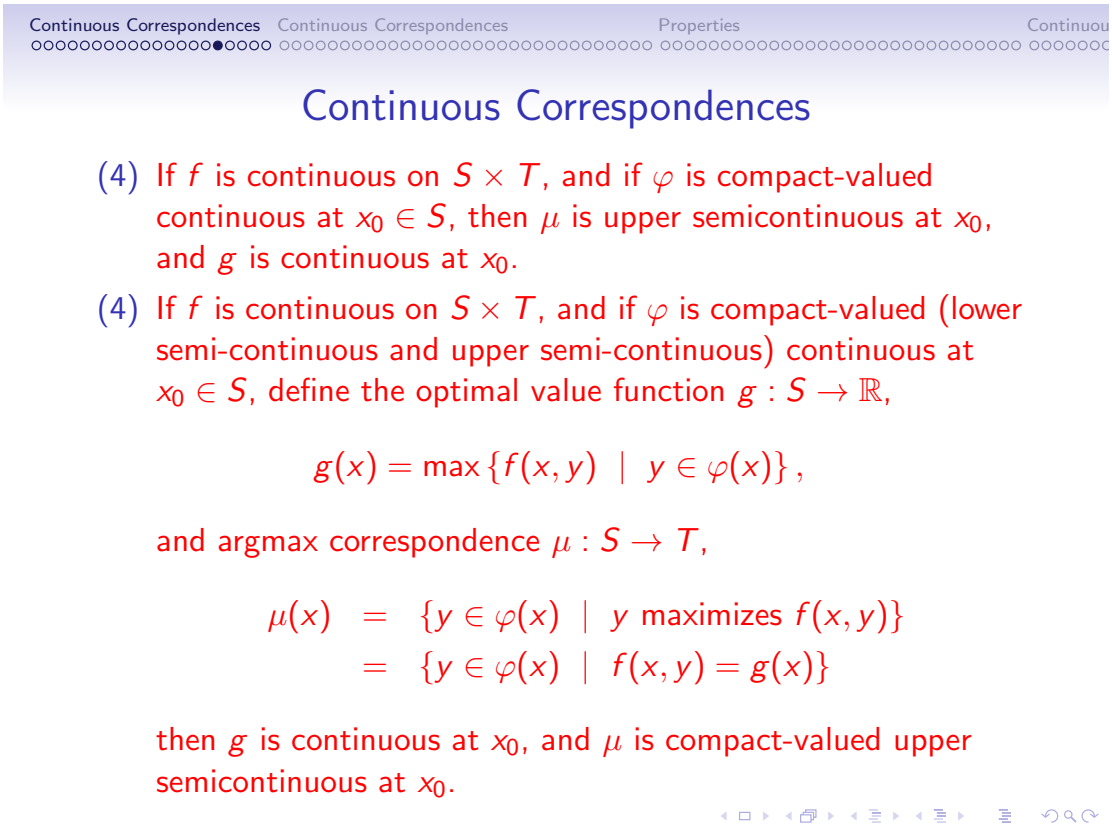
(4) If f is continuous on $S \times T$, and if φ is continuous at $x \in S$, then μ is upper semicontinuous at x, and g is continuous at x.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

若收益 f 连续、可行对应非空紧值且连续，则最优值 g 连续、最优解对应 μ 非空紧值且上半连续。结论没有保证 μ 下半连续。课件括号中的“不要尝试证明 g 连续”是原文的阅读安排，不是本自学讲义的限制；为适应证明考试，19页提供完整证明。

证明路线先记为三步：下半连续把旧的最优行动搬到附近，限制最优值不能骤降；上半连续和紧性把附近的最优行动抽取极限，限制最优值不能骤升；最后将最优等式取极限，保留最优性。

18 原课件第 18 页：把最大值定理的假设写完整



上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页强调紧值性。准确的局部版本是：在 x_0 附近每个 $\varphi(x)$ 非空紧，φ 在 x_0 上、下半连续（开集定义），f 在 $S \times T$ 连续。结论为 g 在 x_0 连续，μ 在 x_0 上半连续且各值非空紧。在本讲前段还可使用共同紧 T 简化抽子列。

紧值只说“每个集合本身紧”；u.s.c. 负责“随着输入接近，集合不会逃走”。两者不能互相代替。课件 $\mu : S \to T$ 更严谨写成 $\mu : S \rightrightarrows T$，因为最优行动可以不唯一。

19 原课件第 19 页：最优值连续的完整证明与原页纠错

Continuous Correspondences

Proof. Fix $x_0 \in S$ and let $x_n \to x_0$. Since φ is lower semi-continuous continuous (l.s.c.) at $x_0 \in S$, for any $y_0 \in \varphi(x_0)$, there exists $(y_n) \subseteq T$ such that $y_n \in \varphi(x_n)$ and $y_n \to y^0$. By the continuity of f, we have $f(x_n, y_n) \to f(x_0, y_0)$. Since $f(x_n, y_n) \leqq g(x_n)$, then

$$f(x_0, y_0) = \lim_{n\to\infty} f(x_n, y_n) \leqq \underline{\lim}_{n\to\infty} g(x_n)$$

and hence $g(x_0) \leqq \underline{\lim}_{n\to\infty} g(x_n)$. Therefore g is lower semi-continuous continuous (l.s.c.) at x_0.

Fix $x_0 \in S$ and let $x_n \to x_0$. Since φ is compact-valued upper semi-continuous continuous (u.s.c.) at $x_0 \in S$, for any sequence $(y_n \in \varphi(x_n))$, the exists a subsequence $(y_{n_k} \in \varphi(x_{n_k}))$ such that $y_{n_k} \to y_0$ and $y^0 \in \varphi(x_0)$. By the continuity of f, we have $f(x_{n_k}, y_{n_k}) \to f(x_0, y_0)$. Since $f(x_{n_k}, y_{n_k}) = g(x_{n_k})$, then

$$g(x_0) = f(x_0, y_0) = \lim_{n\to\infty} f(x_n, y_n) \geqq \overline{\lim}_{n\to\infty} g(x_n).$$

Therefore g is upper semi-continuous continuous (u.s.c.) at x_0.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

固定任意 $x_n \to x_0$，记 $g_0 = g(x_0)$。**先证明下界**。取一个最优点 $y_0 \in \mu(x_0)$。由 φ 下半连续，可选 $z_n \in \varphi(x_n)$ 使 $z_n \to y_0$。连续性给

$$f(x_n, z_n) \to f(x_0, y_0) = g_0, \qquad g(x_n) \geq f(x_n, z_n),$$

故 $\liminf_n g(x_n) \geq g_0$。这里用 $\liminf$，因为还没有证明 $g(x_n)$ 的极限存在。

再证明上界。若 $\limsup_n g(x_n) > g_0$，则存在 $\varepsilon > 0$ 和子列，使 $g(x_{n_k}) \geq g_0 + \varepsilon$。取最优点 $y_k \in \mu(x_{n_k})$，所以 $f(x_{n_k}, y_k) = g(x_{n_k})$。共同紧 T 可抽子列 $y_{k_j} \to y$，闭图给 $y \in \varphi(x_0)$。取极限得

$$f(x_0, y) = \lim_j f(x_{n_{k_j}}, y_{k_j}) \geq g_0 + \varepsilon,$$

但可行性要求 $f(x_0, y) \leq g_0$，矛盾。于是 $\limsup_n g(x_n) \leq g_0$，上下夹住得 $g(x_n) \to g_0$。一般紧值 u.s.c. 情形用第87页的子列结论代替共同紧 T。

纠正： 原页把未知是否存在的 $\lim g(x_n)$ 直接写出来，并把任意可行 y_n 当成最优点；还过早写 $g(x_0) = f(x_0, y_0)$。以上证明逐项补上所需理由。

20 原课件第 20 页：最优解对应为何紧且上半连续

Continuous Correspondences

Proof. For each x_0, f is continuous on $S \times T$, and if φ is compact-valued continuous at $x_0 \in S$, then $\mu(x_0)$ is non-empty and compact-valued from the Extreme Value Theorem.

Let $x_n \to x_0$, $y_n \in \mu(x_n)$. Since φ is compact-valued upper semi-continuous continuous (u.s.c.) at $x_0 \in S$, then there exists $(y_{n_k}) \subseteq T$ such that $y_{n_k} \in \varphi(x_{n_k})$ and $y_{n_k} \to y_0$ and $y_0 \in \varphi(x_0)$. By the continuity of f, we have $f(x_{n_k}, y_{n_k}) \to f(x_0, y_0)$. By the continuity of g, we have $g(x_{n_k}) \to g(x_0)$. Since $f(x_{n_k}, y_{n_k}) = g(x_{n_k})$, then $f(x_0, y_0) = g(x_0)$, so $y_0 \in \mu(x_0)$. Thus μ is upper semicontinuous (u.s.c.) at x_0. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

对固定 x，极值定理给最优点存在。$\mu(x) = \varphi(x) \cap \{y : f(x,y) = g(x)\}$ 是紧集的闭子集，所以紧。接着任取 $x_n \to x_0$、$y_n \in \mu(x_n)$、$y_n \to y$。因为 $\mu(x_n) \subseteq \varphi(x_n)$，闭图给 $y \in \varphi(x_0)$；又

$$f(x_n, y_n) = g(x_n) \implies f(x_0, y) = g(x_0),$$

故 $y \in \mu(x_0)$，最优图闭。

在共同紧 T 中闭图即上半连续。一般情形：若某开集 $G \supseteq \mu(x_0)$ 不能包含附近全部最优点，就可选 $x_n \to x_0$、$y_n \in \mu(x_n) \setminus G$；紧值 u.s.c. 的 φ 给收敛子列，刚才论证使其极限在 $\mu(x_0) \subseteq G$，但各点在闭集 $T \setminus G$ 中，矛盾。只证明“某个好子列存在”而不说明怎样排除坏序列，容易留下逻辑缺口。

21 原课件第 21 页：最优值连续而最优解不下半连续

Continuous Correspondences

Example: Let S and T be the real interval $[0,1]$. Define φ by $\varphi(x) = [0,1]$ for all $x \in S$, and f by $f(x,y) = xy$. For $x \neq 0$, $\mu(x)$ consists of the single element 1; for $x = 0$, $\mu(x) = [0,1]$.

$$\mu(x) = \begin{cases} [0,1], & \text{for} \quad x = 0, \\ \{1\}, & \text{for} \quad x \neq 0. \end{cases}$$

(It will be helpful to draw the graph of μ.) For all x, $g(x) = x$.

$$g(x) = \max\{xy \mid y \in [0,1]\} = x \max\{y \mid y \in [0,1]\} = x.$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$S = T = [0,1]$，$\varphi(x) = [0,1]$，$f(x,y) = xy$。$x > 0$ 时乘以正数保序，所以唯一最优点为1；$x = 0$ 时所有收益都为0，故 $\mu(0) = [0,1]$。$g(x) = x$ 连续。

μ 在0上半连续，因为任何附近最优点的极限都在 $[0,1]$。但不下半连续：取 $x_n = 1/n$ 和目标 $y_0 = 0 \in \mu(0)$，每个 $y_n \in \mu(x_n)$ 都只能等于1，不能趋于0。这个反例说明最大值定理的结论已经是恰当强度，不能擅自把 u.s.c. 改为连续。

22 原课件第 22 页：紧值域假设什么时候能替换



I. Throughout this section, the assumption that T is compact has been made; it is essential in several respects. In applications, when the set T is not compact, one may still be able to replace T by some compact set without changing the problem, and thus to use the above results.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

若行动空间原本是 $\mathbb{R}^L$，可以证明所有相关可行点（或所有相关最优点）位于某个共同紧集，再在该集上运用前面的定理。例如收入 w 局部有上界、每个商品价格 p_i 局部有共同正下界 $c>0$，预算约束推出 $0 \le y_i \le w/c$，从而有统一有界容器。价格允许趋于零时这个论证会失效。不可直接把不紧的 T 写成紧而不证明截断不会丢掉所需选择。

23 原课件第 23 页：上下半连续对应的邻域含义

Semi-continuous Mappings

In this section, S denotes a subset of $\mathbb{R}^K$, T a subset of $\mathbb{R}^L$, and a φ correspondence from S to T, that is $\varphi : S \rightrightarrows T$. For correspondences there are two natural notions of inverse.

(1) The lower (or weak) inverse of $E \subseteq T$ under $\varphi : S \rightrightarrows T$, denoted $\varphi^-(E)$, is defined by

$$\varphi^-(E) = \{x \in S : \varphi(x) \cap E \neq \varnothing\}.$$

(2) The upper (or strong) inverse of $E \subseteq T$ under $\varphi : S \rightrightarrows T$, denoted $\varphi^+(E)$, is defined by

$$\varphi^+(E) = \{x \in S : \varphi(x) \subseteq E\}.$$

Remark. For a single-valued mapping $\varphi : S \rightrightarrows T$, define

$$\varphi^{-1}(y) = \{x \in S : y \in \varphi(x)\} = \{x \in S : y = \varphi(x)\}.$$

Note that $\varphi^{-1}(y) = \varphi^-(\{y\})$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

下半连续：每个与 $\Phi(x)$ 相交的开集 V，在附近每个输入的值集仍与 V 相交。上半连续：每个包含整个 $\Phi(x)$ 的开集 V，附近全部值集都包含于 V。前者保证旧选择可近似保留，后者禁止新选择逃出旧值集的开邻域。对单点值，两定义均化为普通函数连续；这里的上、下不是实值函数大小方向。

24 原课件第 24 页：上下半连续对应的邻域含义

Semi-continuous Mappings

Let φ be a correspondence of a topological space S into a topological space T and let x_0 be a point of S.

- We say that φ is lower semi-continuous (l.s.c.) at x_0 if for each open set G meeting $\varphi(x_0)$ $[\varphi(x_0) \cap G \neq \varnothing]$ there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \cap G \neq \varnothing,$$

  i.e. $\varphi(U(x_0)) \cap G \neq \varnothing$.

- We say that φ is upper semi-continuous (u.s.c.) at x_0 if for each open set G containing $\varphi(x_0)$ $[\varphi(x_0) \subseteq G]$ there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \subseteq G,$$

  i.e. $\varphi(U(x_0)) \subseteq G$.

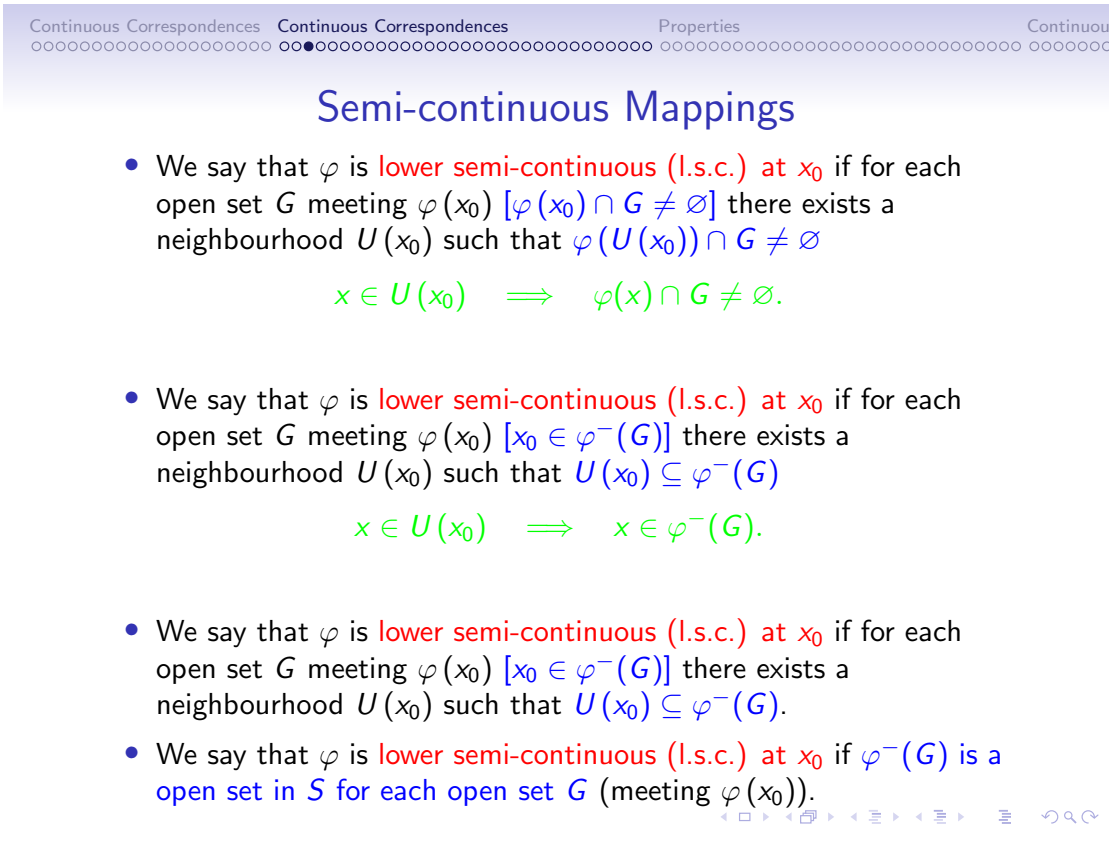
We say that the mapping φ is continuous at x_0 if it is both lower and upper semi-continuous at x_0.

$$\varphi^-(E) = \{x \in S \mid \varphi(x) \cap E \neq \varnothing\}$$
$$\varphi^+(E) = \{x \in S \mid \varphi(x) \subset E\}$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“上下半连续对应的邻域含义”。完整定义、条件和论证见本讲原页第 23 页下方。

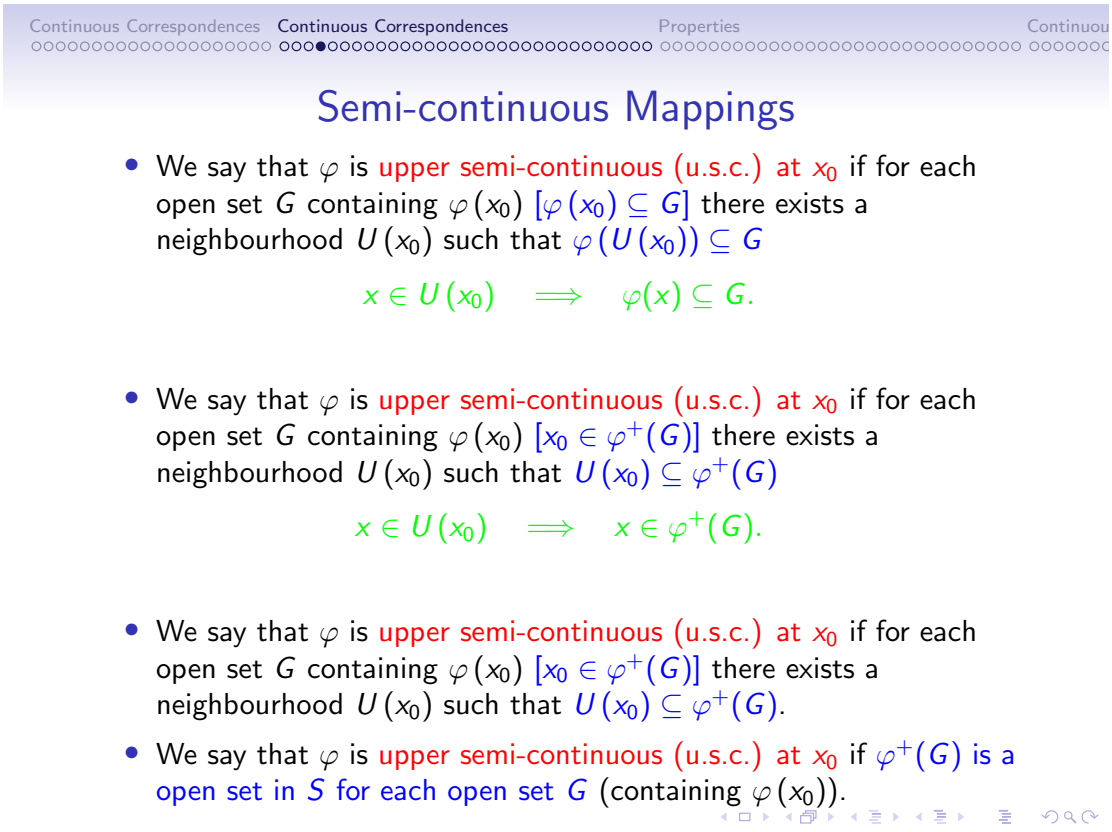
25 原课件第 25 页：上下半连续对应的邻域含义



上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“上下半连续对应的邻域含义”。完整定义、条件和论证见本讲原页第 23 页下方。

26 原课件第 26 页：上下半连续对应的邻域含义



上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“上下半连续对应的邻域含义”。完整定义、条件和论证见本讲原页第 23 页下方。

27 原课件第 27 页：上下半连续对应的邻域含义

Semi-continuous Mappings

If φ is a single-valued mapping, the definition given above for lower semi-continuity coincides with the ordinary definition of continuity; the same is true for upper semi-continuity.

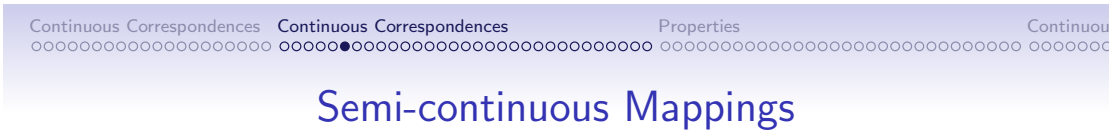
- We say that φ is lower semi-continuous (l.s.c.) in S if it is lower semi-continuous (l.s.c.) at each point of S, that is, $\varphi^{-}(G)$ is a open set in S for each open set G.
- We say that φ is upper semi-continuous (u.s.c.) in S if it is upper semi-continuous (u.s.c.) at each point of S, that is, $\varphi^{+}(G)$ is a open set in S for each open set G.

We say that the correspondence φ is continuous in S if it is both lower and upper semi-continuous in S.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“上下半连续对应的邻域含义”。完整定义、条件和论证见本讲原页第 23 页下方。

28 原课件第 28 页：逆像刻画的证明



Theorem 1.1. $\varphi : S \rightrightarrows T$ is a correspondence, then the following three statements are equivalent.

(1) φ is lower semi-continuous (l.s.c.).

(2) $\varphi^{-}(G)$ is a open set in S for any open set $G \subseteq T$.

(3) $\varphi^{+}(F)$ is a closed set in S for any closed set $F \subseteq T$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

下逆像 $\Phi^{-}(V) = \{x : \Phi(x) \cap V \neq \varnothing\}$。下半连续当且仅开集的下逆像开：若某点属于该逆像，定义给整个邻域也属于它；反向取逆像本身为邻域。上逆像 $\Phi^{+}(V) = \{x : \Phi(x) \subseteq V\}$，同样论证得到上半连续的开逆像判据。非空值下的补集对偶把两种开集判据换成闭集判据；允许空值时应先限制到有效定义域。

29 原课件第 29 页：逆像刻画的证明

Semi-continuous Mappings

Proof of (1) $\Longrightarrow$ (2) For any open set $G \subseteq T$ we want to prove that $\varphi^{-}(G)$ is a open set in S. If $x_0 \in \varphi^{-}(G)$, then $\varphi(x_0) \cap G \neq \varnothing$, and so there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \cap G \neq \varnothing,$$

i.e. $\varphi(U(x_0)) \cap G \neq \varnothing$, whence

$$x \in U(x_0) \quad \Longrightarrow \quad x \in \varphi^{-}(G),$$

i.e. $U(x_0) \subseteq \varphi^{-}(G)$. Therefore $\varphi^{-}(G)$ is open.

Proof of (2) $\Longrightarrow$ (1) Suppose now that $\varphi^{-}(G)$ is an open set for each open set G in T. If $x_0 \in \varphi^{-}(G)$, then there exists a neighbourhood $U(x_0)$ such that $U(x_0) \subseteq \varphi^{-}(G)$, that is, if G meets $\varphi(x_0)$ $[\varphi(x_0) \cap G \neq \varnothing]$, then there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad x \in \varphi^{-}(G),$$

hence

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \cap G \neq \varnothing,$$

i.e. $\varphi(U(x_0)) \cap G \neq \varnothing$, thus φ is lower semi-continuous (l.s.c.) at x_0.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“逆像刻画的证明”。完整定义、条件和论证见本讲原页第 28 页下方。

30 原课件第 30 页：逆像刻画的证明

Semi-continuous Mappings

Proof of (2) $\Longrightarrow$ (3) For any closed set $F \subseteq T$, $G = \mathbf{C}_T F$ is an open set.

$$\begin{aligned}
\varphi^-(G) &= \{x \in S \mid \varphi(x) \cap G \neq \varnothing\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \cap G = \varnothing\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \subseteq \mathbf{C}_T G\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \subseteq F\} \\
&= \mathbf{C}_S\left(\varphi^+(F)\right)
\end{aligned}$$

is open, that is, $\varphi^+(F)$ is closed.

Proof of (3) $\Longrightarrow$ (2) For any open set $G \subseteq T$, $F = \mathbf{C}_T G$ is a closed set.

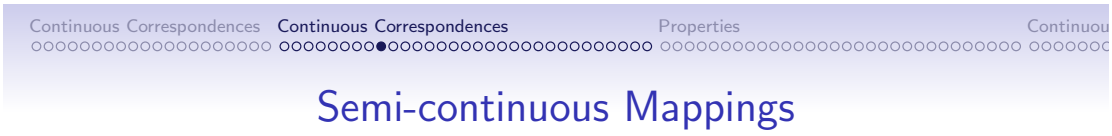
$$\begin{aligned}
\varphi^+(F) &= \{x \in S \mid \varphi(x) \subseteq F\} \\
&= \{x \in S \mid \varphi(x) \subseteq \mathbf{C}_T G\} \\
&= \{x \in S \mid \varphi(x) \cap G = \varnothing\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \cap G \neq \varnothing\} \\
&= \mathbf{C}_S\left(\varphi^-(G)\right)
\end{aligned}$$

is closed, that is, $\varphi^-(G)$ is open. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“逆像刻画的证明”。完整定义、条件和论证见本讲原页第 28 页下方。

31 原课件第 31 页：逆像刻画的证明



Theorem 1.2. $\varphi : S \rightrightarrows T$ is a correspondence, then the following three statements are equivalent.

(1) φ is upper semi-continuous (u.s.c.).

(2) $\varphi^{+}(G)$ is a open set in S for any open set $G \subseteq T$.

(3) $\varphi^{-}(F)$ is a closed set in S for any closed set $F \subseteq T$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“逆像刻画的证明”。完整定义、条件和论证见本讲原页第 28 页下方。

32 原课件第 32 页：逆像刻画的证明

Semi-continuous Mappings

Proof of (1) $\Longrightarrow$ (2) For any open set $G \subseteq T$ we want to prove that $\varphi^+(G)$ is a open set in S. If $x_0 \in \varphi^+(G)$, then $\varphi(x_0) \subseteq G$, and so there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \subseteq G,$$

i.e. $\varphi(U(x_0)) \subseteq G$, whence

$$x \in U(x_0) \quad \Longrightarrow \quad x \in \varphi^+(G),$$

i.e. $U(x_0) \subseteq \varphi^+(G)$. Therefore $\varphi^+(G)$ is a neighbourhood of each of its points and so is open.

Proof of (2) $\Longrightarrow$ (1) Suppose now that $\varphi^+(G)$ is an open set for each open set G in T. If $x_0 \in \varphi^+(G)$, then there exists a neighbourhood $U(x_0)$ such that $U(x_0) \subseteq \varphi^+(G)$, that is, if G contains $\varphi(x_0)$ $[\varphi(x_0) \subseteq G]$, then there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad x \in \varphi^+(G),$$

hence

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \subseteq G,$$

i.e. $\varphi(U(x_0)) \subseteq G$, thus φ is upper semi-continuous (u.s.c.) at x_0.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“逆像刻画的证明”。完整定义、条件和论证见本讲原页第 28 页下方。

33 原课件第 33 页：逆像刻画的证明

Semi-continuous Mappings

Proof of (2) $\Longrightarrow$ (3) For any closed set $F \subseteq T$, $G = \mathbf{C}_T F$ is an open set.

$$\begin{aligned}
\varphi^+(G) &= \{x \in S \mid \varphi(x) \subseteq G\} \\
&= \{x \in S \mid \varphi(x) \subseteq \mathbf{C}_T F\} \\
&= \{x \in S \mid \varphi(x) \cap F = \varnothing\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \cap F \neq \varnothing\} \\
&= \mathbf{C}_S\left(\varphi^-(F)\right)
\end{aligned}$$

is open, that is, $\varphi^-(F)$ is closed.

Proof of (3) $\Longrightarrow$ (2) For any open set $G \subseteq T$, $F = \mathbf{C}_T G$ is a closed set.

$$\begin{aligned}
\varphi^-(F) &= \{x \in S \mid \varphi(x) \cap F \neq \varnothing\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \cap F = \varnothing\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \subseteq \mathbf{C}_T F\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \subseteq G\} \\
&= \mathbf{C}_S\left(\varphi^+(G)\right)
\end{aligned}$$

is closed, that is, $\varphi^+(G)$ is open. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“逆像刻画的证明”。完整定义、条件和论证见本讲原页第 28 页下方。

34 原课件第 34 页：紧值上半连续对应的紧像

Semi-continuous Mappings

Upper semi-continuous (u.s.c.) correspondences of S into T have the following fundamental property:

Theorem 1.3. If $\varphi : S \to T$ is a compact-valued u.s.c. correspondence, then the image $\varphi(K)$ of a compact subset K of S is also compact.

Proof. Let $\{G_\lambda \mid \lambda \in \Lambda\}$ be an open covering of $\varphi(K)$. If $x \in K$, the set $\varphi(x)$, which is compact, can be covered by a finite number of the $G_\lambda(x)$ for $\lambda \in \Lambda$, $\varphi(x) \subseteq \bigcup_{\lambda\in\Lambda} G_\lambda(x)$; let G_x denote the union of the sets in such a finite family, $G_x = \bigcup_{\lambda\in\Lambda} G_\lambda(x)$, then $\varphi(K) \subseteq \bigcup_{x\in K} \varphi(x) \subseteq \bigcup_{x\in K} G_x$. Then $\{\varphi^+(G_x) \mid x \in K\}$ is an open covering of K and so it contains a finite covering $\varphi^+(G_{x_1}), \cdots, \varphi^+(G_{x_n}), \cdots, \varphi^+(G_{x_N})$. $K \subseteq \bigcup_{n=1}^{N} \varphi^+(G_{x_n})$. The sets $G_{x_1}, \cdots, G_{x_n}, \cdots, G_{x_N}$ cover $\varphi(K)$ and so $\varphi(K)$ can be covered by a finite number of the G_{x_n} for $n = 1, \cdots, N$. That is, $\varphi(K) \subseteq \bigcup_{n=1}^{N} G_{x_n}$.
□

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

设 K 紧且 Φ 紧值上半连续。给 $\Phi(K)$ 的任意开覆盖，对每个 $x \in K$ 先有限覆盖紧集 $\Phi(x)$，把对应有限个开集的并记作 V_x。上半连续给 U_x 使 $\Phi(U_x) \subset V_x$。再用 K 紧，从 U_x 取有限子覆盖。两次有限选择的总开集数仍有限，覆盖 $\Phi(K)$。两个紧性分别作用于“每个值”和“输入集”。

35 原课件第 35 页：闭对应与闭图

Semi-continuous Mappings

In addition to the two types of semi-continuity, it is sometimes convenient to consider a third topological property.

Definition 1.2. (Closed Correspondence) We say that φ is a closed correspondence of S into T if whenever $x_0 \in S$, $y_0 \in T$, $y_0 \notin \varphi(x_0)$ there exist two neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$ implies $\varphi(x) \cap V(y_0) = \varnothing$.

Consider the graphical representation $\sum_{x \in S} \varphi(x)$ of φ in $S \times T$;

$$G(\varphi) \equiv \sum_{x \in S} \varphi(x) = \{(x, y) \in S \times T \mid y \in \varphi(x)\}.$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

闭图指若 $x_n \to x, y_n \to y, y_n \in \Phi(x_n)$，则 $y \in \Phi(x)$。等价于图在乘积空间闭；也等价于图外每个 (x, y) 可找到乘积开邻域与图不交。固定 x 用常数输入得闭图蕴含闭值。连续 $f(x, y)$ 的弱不等式定义对应有闭图，因为不等式保极限。任意闭图对应的逐点交为图的任意闭交，有限并为有限闭并，有限积由坐标判据闭。闭值本身不推出闭图，例如单值跳跃函数。

36 原课件第 36 页：闭对应与闭图

Semi-continuous Mappings

- We say that φ is a closed correspondence of S into T if whenever $x_0 \in S$, $y_0 \in T$, $y_0 \notin \varphi(x_0)$ there exist two neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$ implies $\varphi(x) \cap V(y_0) = \varnothing$.
- We say that φ is a closed correspondence of S into T if whenever $x_0 \in S$, $y_0 \in T$, $y_0 \notin \varphi(x_0)$ there exist two neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$ and $y \in V(y_0)$ implies $y \notin \varphi(x)$.
- We say that φ is a closed correspondence of S into T if whenever $x_0 \in S$, $y_0 \in T$, $(x_0, y_0) \in \mathbf{C}(G(\varphi))$ there exist two neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$ and $y \in V(y_0)$ implies $(x, y) \in \mathbf{C}(G(\varphi))$.
- We say that φ is a closed correspondence of S into T if whenever $x_0 \in S$, $y_0 \in T$, $(x_0, y_0) \in \mathbf{C}(G(\varphi))$ there exist two neighbourhoods $U(x_0)$ and $V(y_0)$ such that $U(x_0) \times V(y_0) \subseteq \mathbf{C}(G(\varphi))$.
- We say that φ is a closed correspondence of S into T if $\mathbf{C}(G(\varphi))$ is an open set in $S \times T$.
- We say that φ is a closed correspondence of S into T if $G(\varphi)$ is a closed set in $S \times T$.

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本页继续“闭对应与闭图”。完整定义、条件和论证见本讲原页第 35 页下方。

37 原课件第 37 页：闭对应与闭图

Semi-continuous Mappings

Proposition 1.1. Let S and T be two metric spaces. A mapping $\varphi : S \rightrightarrows T$ is a closed correspondence if and only if $G(\varphi)$ is a closed set in $S \times T$.

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38 原课件第 38 页：闭对应与闭图

Semi-continuous Mappings

- We say that φ is a closed correspondence of S into T if whenever $x_0 \in S$, $y_0 \in T$, $y_0 \notin \varphi(x_0)$ there exist two neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$ implies $\varphi(x) \cap V(y_0) = \varnothing$.
- We say that φ is a closed correspondence of S into T if whenever $x_0 \in S$, $y_0 \in T$, for any two neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$ implies $\varphi(x) \cap V(y_0) \neq \varnothing$ [$y \in \varphi(x) \cap V(y_0)$ if and only if $y \in \varphi(x)$ and $y \in V(y_0)$], then $y_0 \in \varphi(x_0)$.
- We say that φ is a closed correspondence of S into T if whenever $x_0 \in S$, $y_0 \in T$, for any two neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$, $y \in V(y_0)$, and $y \in \varphi(x)$, then $y_0 \in \varphi(x_0)$.
- We say that φ is a closed correspondence of S into T if whenever $x_n \longrightarrow x_0$, $y_n \longrightarrow y_0$, $y_n \in \varphi(x_n)$, then $y_0 \in \varphi(x_0)$.

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39 原课件第 39 页：闭对应与闭图

Semi-continuous Mappings

Proposition 1.2. Let S and T be two metric spaces. A mapping $\varphi : S \rightrightarrows T$ is a closed correspondence if and only if whenever $x_n \longrightarrow x_0$, $y_n \longrightarrow y_0$, $y_n \in \varphi(x_n)$, then $y_0 \in \varphi(x_0)$.

Theorem 1.4. $G(\varphi)$ is a closed set in $S \times T$ if and only if whenever $x_n \longrightarrow x_0$, $y_n \longrightarrow y_0$, $y_n \in \varphi(x_n)$, then $y_0 \in \varphi(x_0)$.

Proof. $G(\varphi)$ is a closed set in $S \times T$ if and only if whenever $(x_n, y_n) \longrightarrow (x_0, y_0)$ and $(x_n, y_n) \in G(\varphi)$, then $(x_0, y_0) \in G(\varphi)$ if and only if whenever $x_n \longrightarrow x_0$, $y_n \longrightarrow y_0$, $y_n \in \varphi(x_n)$, then $y_0 \in \varphi(x_0)$. □

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40 原课件第 40 页：闭对应与闭图

Properties of Semi-continuous Mappings

Proposition 1.3. If every $\varphi_n : S \rightrightarrows T_n$ is a closed correspondence for $n = 1, \cdots, N$, then the Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n : S \rightrightarrows \prod_{n=1}^{N} T_n$ is a closed correspondence.

Proof 1. If $\left(y_1^0, \cdots, y_N^0\right) = y^0 \notin \varphi\left(x^0\right) = \prod_{n=1}^{N} \varphi_n\left(x^0\right)$, then there exists $n_0 = 1, \cdots, N$ such that $y_{n_0}^0 \notin \varphi_{n_0}\left(x^0\right)$. Therefore there exist neighbourhoods $U_{n_0}\left(x^0\right)$ and $V_{n_0}\left(y_{n_0}^0\right)$ such that $x \in U_{n_0}\left(x^0\right)$ implies $\varphi_{n_0}(x) \cap V_{n_0}\left(y_{n_0}^0\right) = \varnothing$, that is, $\varphi_{n_0}\left(U_{n_0}\left(x^0\right)\right) \cap V_{n_0}\left(y_{n_0}^0\right) = \varnothing$. Take $U\left(x^0\right) = \bigcap_{n=1}^{N} U_n\left(x^0\right)$, then $x \in U\left(x_0\right)$ implies

$$\varphi_{n_0}\left(U\left(x^0\right)\right) \cap V_{n_0}\left(y_{n_0}^0\right) \subseteq \varphi_{n_0}\left(U_{n_0}\left(x^0\right)\right) \cap V_{n_0}\left(y_{n_0}^0\right) = \varnothing$$

and hence $\varphi_{n_0}\left(U\left(x^0\right)\right) \cap V_{n_0}\left(y_{n_0}^0\right) = \varnothing$. Take $V\left(y^0\right) = \prod_{n=1}^{N} V_n\left(y_n^0\right)$, then

$$\varphi\left(U\left(x^0\right)\right) \cap V\left(y^0\right) = \prod_{n=1}^{N}\left[\varphi_n\left(U\left(x^0\right)\right) \cap V_n\left(y_n^0\right)\right] = \varnothing;$$

whence φ is a closed mapping. □

Proof 2. If $x^q \to x^0$, $\left(y_1^q, \cdots, y_N^q\right) = y^q \to y^0 = \left(y_1^0, \cdots, y_N^0\right)$, and $\left(y_1^q, \cdots, y_N^q\right) = y^q \in \varphi\left(x^q\right) = \prod_{n=1}^{N} \varphi_n\left(x^q\right)$, then $y_n^q \to y_n^0$ and $y_n^q \in \varphi_n\left(x^q\right)$ for $n = 1, \cdots, N$. If every φ_n is closed at the point x_0 of S, then $y_n^0 \in \varphi_n\left(x^0\right)$ for $n = 1, \cdots, N$, and hence $y^0 = \left(y_1^0, \cdots, y_N^0\right) \in \prod_{n=1}^{N} \varphi_n\left(x^0\right) = \varphi\left(x^0\right)$. Thus, $\varphi = \prod_{n=1}^{N} \varphi_n$ is closed at the point x_0 of S. □

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41 原课件第 41 页：闭对应与闭图

Semi-continuous Mappings

Proposition 1.4. Let S and T be two metric spaces. If $\varphi : S \rightrightarrows T$ is a closed correspondence at x, then the set $\varphi(x)$ is closed in T.

Proof: Let y_n be a sequence in $\varphi(x)$ with $y_n \to y$. Define $x_n = x$ for all n, so $x_n \to x$ and $y_n \in \varphi(x_n)$. Since φ is a closed correspondence at x, we have $y \in \varphi(x)$. This shows that $\varphi(x)$ is a closed set. □

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42 原课件第 42 页：闭对应与闭图

Semi-continuous Mappings

Example 1.1. If f is a continuous numerical function in $S \times T$, the correspondence $\varphi : S \rightrightarrows T$ defined by $\varphi(x) = \{y \in T \mid f(x, y) \leqq 0\}$ is a closed correspondence of S into T, for, in $S \times T$, the graphical representation

$$G(\varphi) = \{(x, y) \in S \times T \mid f(x, y) \leqq 0\}$$

is a closed set.

In particular, if λ is a continuous numerical function in a metric space (X, d), the mapping

$$\varphi(x) = B_{\lambda(x)}(x) = \{y \in X \mid d(x, y) - \lambda(x) \leqq 0\}$$

is closed.

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43 原课件第 43 页：闭对应与闭图

Semi-continuous Mappings

Theorem 1.5. If $\{\varphi_\lambda \mid \lambda \in \Lambda\}$ is a family of closed correspondences of S into T, then $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ is also a closed correspondence.

Proof 1. If $y_0 \notin \varphi(x_0) = \bigcap_{\lambda\in\Lambda} \varphi_\lambda(x_0)$, then there exists an index $\lambda_0 \in$ such that

$$y_0 \notin \varphi_{\lambda_0}(x_0).$$

Therefore there exist neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$ implies $\varphi_{\lambda_0}(x) \cap V(y_0) = \varnothing$, that is,

$$\varphi_{\lambda_0}(U(x_0)) \cap V(y_0) = \varnothing$$

and so $x \in U(x_0)$ implies $\varphi(x) \cap V(y_0) = \bigcap_{\lambda\in\Lambda} [\varphi_\lambda(x) \cap V(y_0)] = \varnothing$, that is,

$$\varphi(U(x_0)) \cap V(y_0) = \varnothing;$$

whence φ is a closed mapping. □

Proof 2. If $x_n \to x_0$, $y_n \to y_0$, and $y_n \in \varphi(x_n) = \bigcap_{\lambda\in\Lambda} \varphi_\lambda(x_n)$, then $y_n \in \varphi_\lambda(x_n)$ for $\lambda \in \Lambda$. If every φ_λ is closed at the point x_0 of S, then $y_0 \in \varphi_\lambda(x_0)$ for $\lambda \in \Lambda$, and hence $y_0 \in \bigcap_{\lambda\in\Lambda} \varphi_\lambda(x_0) = \varphi(x_0)$. Thus, $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ is closed at the point x_0 of S. □

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44 原课件第 44 页：闭对应与闭图

Properties of Semi-continuous Mappings

Proposition 1.5. If every $\varphi_n : S \rightrightarrows T_n$ is a closed correspondence for $n = 1, \cdots, N$, then the union $\varphi = \bigcup_{n=1}^{N} \varphi_n$ is a closed correspondence.

Proof 1. If $y_0 \notin \varphi(x_0) = \bigcup_{n=1}^{N} \varphi_n(x_0)$, then, for all $n = 1, \cdots, N$,

$$y_0 \notin \varphi_n(x_0).$$

Therefore there exist neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$ implies $\varphi_n(x) \cap V(y_0) = \varnothing$, that is,

$$\varphi_n(U(x_0)) \cap V(y_0) = \varnothing$$

and so $x \in U(x_0)$ implies $\varphi(x) \cap V(y_0) = \bigcup_{n=1}^{N} [\varphi_n(x) \cap V(y_0)] = \varnothing$, that is,

$$\varphi(U(x_0)) \cap V(y_0) = \bigcup_{n=1}^{N} [\varphi_n(U(x_0)) \cap V(y_0)] = \varnothing;$$

whence φ is a closed mapping. □

Proof 2. If $x^q \to x^0$, $y^q \to y^0$, and $y^q \in \varphi(x^q) = \bigcup_{n=1}^{N} \varphi_n(x^q)$, then there exists $n_0 \in \{1, \cdots, N\}$ such that $y^q \in \varphi_{n_0}(x^q)$. If every φ_n is closed at the point x_0 of S, then $y^0 \in \varphi_{n_0}\left(x^0\right)$, and hence $y^0 \in \bigcup_{n=1}^{N} \varphi_n\left(x^0\right) = \varphi\left(x^0\right)$. Thus, $\varphi = \bigcup_{n=1}^{N} \varphi_n$ is closed at the point x_0 of S. □

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45 原课件第 45 页：闭图与上半连续之间的紧性桥梁

Semi-continuous Mappings

Theorem 1.6. Every compact-valued u.s.c. correspondence is closed.

Proof. Let φ be a upper semi-continuous (u.s.c.) correspondence of S into T and suppose that $y_0 \notin \varphi(x_0)$. Since $\varphi(x_0)$ is compact, there exists an open set G in T containing $\varphi(x_0)$ and a neighbourhood $V(y_0)$ such that

$$G \cap V(y_0) = \varnothing.$$

Since φ is a upper semi-continuous (u.s.c.) correspondence, there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \Longrightarrow \varphi(x) \subseteq G.$$

Then

$$x \in U(x_0) \Longrightarrow \varphi(x) \cap V(y_0) = \varnothing$$

and φ is therefore closed. □

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紧值上半连续对应有闭图：若 $y \notin \Phi(x)$，紧值闭且 $\operatorname{dist}(y, \Phi(x)) > 0$，取分离开邻域，再用上半连续，附近图不能靠近 (x, y)。若 Ψ 闭图，Φ 紧值上半连续，则 $\Phi \cap \Psi$ 紧值上半连续。证明反证：若有 $x_n \to x$ 和交值 y_n 逃出包含交值的开集 G，紧像性质给子列 $y_n \to y$，两图闭使 y 属于交值，开性使子列最终进 G，矛盾。值域空间紧时，闭图亦推出上半连续：逃出开集的选择总能抽取收敛子列。

46 原课件第 46 页：闭图与上半连续之间的紧性桥梁



Theorem 1.7. If φ_1 is a closed mapping of S into T and φ_2 is a compact-valued u.s.c. correspondence of S into T, then the correspondence $\varphi = \varphi_1 \cap \varphi_2$ is compact-valued u.s.c.

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47 原课件第 47 页：闭图与上半连续之间的紧性桥梁

Semi-continuous Mappings

Proof. For each x, the set $\varphi(x)$ is compact, because it is a closed set contained in a compact set $\varphi_2(x)$. Thus φ is compact-valued.
Let G be an open set such that $\varphi(x_0) = \varphi_1(x_0) \cap \varphi_2(x_0) \subseteq G$. We shall prove that there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \Longrightarrow \varphi(x) \subseteq G$$

that is, $\varphi(U(x_0)) \subseteq G$.
(1) If $\varphi_2(x_0) \subseteq G$, there is nothing to prove. φ_2 is upper semi-continuous (u.s.c.), there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \Longrightarrow \varphi_2(x) \subseteq G$$

that is, $\varphi_2(U(x_0)) \subseteq G$ and hence $\varphi(U(x_0)) \subseteq \varphi_2(U(x_0)) \subseteq G$.
(2) Suppose that $\varphi_2(x_0) \cap (\mathbf{C}G) = K \neq \varnothing$. Let y be a point of K, $y \notin \varphi_1(x_0)$; then there exist neighbourhoods $V(y)$ and $U_y(x_0)$ such that

$$x \in U_y(x_0) \Longrightarrow \varphi_1(x) \cap V(y) = \varnothing,$$

that is, $\varphi_1(U_y(x_0)) \cap V(y) = \varnothing$.

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48 原课件第 48 页：闭图与上半连续之间的紧性桥梁

Semi-continuous Mappings

Since the set K is compact, there exist elements $y_1, y_2, \cdots, y_N$ in K such that $V(y_1), V(y_2), \cdots, V(y_N)$ cover K. Write $K \subseteq \bigcup_{n=1}^{N} V(y_n) \equiv V(K)$, then $\varphi_2(x_0) \subseteq G \cup V(K)$.
If φ_2 is upper semi-continuous (u.s.c.), then there exists a neighbourhood $U'(x_0)$ such that

$$x \in U'(x_0) \Longrightarrow \varphi_2(x) \in G \cup V(K),$$

that is, $\varphi_2(U'(x_0)) \subseteq G \cup V(K)$.
Putting $U(x_0) = U_{y_1}(x_0) \cap U_{y_2}(x_0) \cap \cdots \cap U_{y_N}(x_0) \cap U'(x_0)$, we have

$$\varphi_1(U(x_0)) \cap V(y_n) \subseteq \varphi_1(U_{y_n}(x_0)) \cap V(y_n) = \varnothing,$$

that is, $\varphi_1(U(x_0)) \cap V(y_n) = \varnothing$, and hence

$$\varphi_1(U(x_0)) \cap V(K) = \varphi_1(U(x_0)) \cap \bigcup_{n=1}^{N} V(y_n) = \bigcup_{n=1}^{N} [\varphi_1(U(x_0)) \cap V(y_n)] = \varnothing$$

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49 原课件第 49 页：闭图与上半连续之间的紧性桥梁

Semi-continuous Mappings

Therefore

$$\begin{cases} (\varphi_1 \cap \varphi_2)(U(x_0)) \cap V(K) \subseteq \varphi_1(U(x_0)) \cap V(K) = \varnothing, \\ (\varphi_1 \cap \varphi_2)(U(x_0)) \subseteq \varphi_2(U(x_0)) \subseteq G \cup V(K), \end{cases}$$

whence

$$\varphi(U(x_0)) = (\varphi_1 \cap \varphi_2)(U(x_0)) \subseteq G,$$

that is,

$$x \in U(x_0) \Longrightarrow (\varphi_1 \cap \varphi_2)(x) \subseteq G.$$

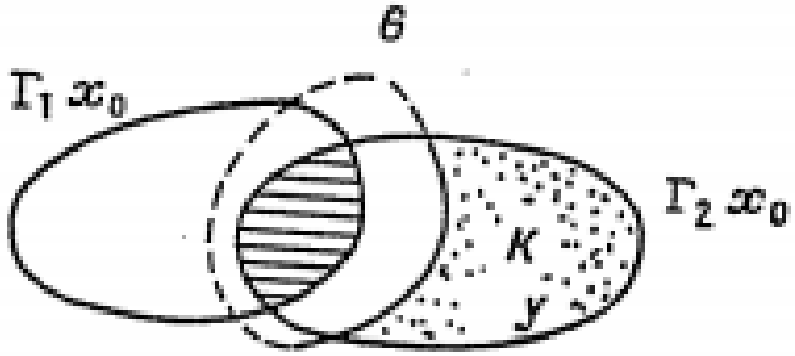
Therefore φ is upper semi-continuous (u.s.c.). $\square$

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50 原课件第 50 页：闭图与上半连续之间的紧性桥梁

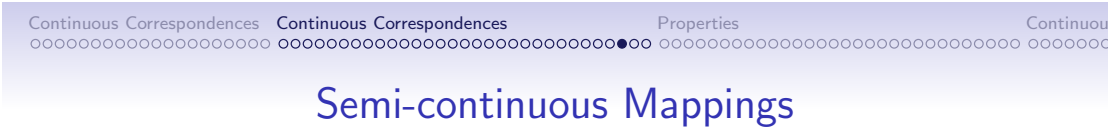
Semi-continuous Mappings



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51 原课件第 51 页：闭图与上半连续之间的紧性桥梁



Corollary 1.1. If T is a compact space, a correspondence $\varphi : S \rightrightarrows T$ is closed-valued, then the correspondence φ is closed if and only if it is upper semi-continuous (u.s.c.).

Proof. If T is a compact space and φ is a closed-valued mapping, $\varphi(x)$ is a closed set for each $x \in S$, thus $\varphi(x)$ is a compact set and hence φ is compact-valued. If φ is u.s.c., then it is closed. Conversely, $\Delta : S \to \{T\}$ is the mapping such that $\Delta(x) = T$ for each x, Δ is a compact-valued u.s.c. correspondence. If φ is closed, then $\varphi = \varphi \cap \Delta$ is u.s.c. $\square$

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52 原课件第 52 页：紧空间上的不变紧集

Semi-continuous Mappings

Theorem 1.8. If S is a compact space and φ is a upper semi-continuous (u.s.c.) correspondence of S into itself such that, for each x, $\varphi(x) \neq \varnothing$, then there exists a non-empty compact subset K of S such that $\varphi(K) = K$.

Proof. The sets $S, \varphi(S), \varphi^2(S), \cdots$ form a sequence of non-empty compact sets; if any of these is equal to its successor, it is a set of the required kind and so the desired result is proved. We therefore assume that the sets are distinct; since $S \supseteq \varphi(S)$, we have $\varphi(S) \supseteq \varphi^2(S)$, $\varphi^2(S) \supseteq \varphi^3(S)$ and so on. Thus the sequence of sets $\varphi^n(S)$ is decreasing. By the finite intersection axiom, we have

$$K = \bigcap_{n=1}^{\infty} \varphi^n(S) \neq \varnothing.$$

For each n, we have $K \subseteq \varphi^{n-1}(S)$ and so $\varphi(K) \subseteq \varphi^n(S)$, whence $\varphi(K) \subseteq K$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

假设非空紧 S、非空紧值上半连续自对应。令 $K_n = \Phi^n(S)$，紧像定理使它们非空紧，单调性使 $K_{n+1} \subseteq K_n$，故交 K 非空紧。单调性给 $\Phi(K) \subseteq K$。对 $a \in K$，可取 $x_n \in K_n$ 使 $a \in \Phi(x_n)$，抽子列 $x_n \to x$，套集闭性给 $x \in K$，闭图给 $a \in \Phi(x)$。故 $K = \Phi(K)$。这不保证某点 $x \in \Phi(x)$，不要把不变集合误当不动点。

53 原课件第 53 页：紧空间上的不变紧集

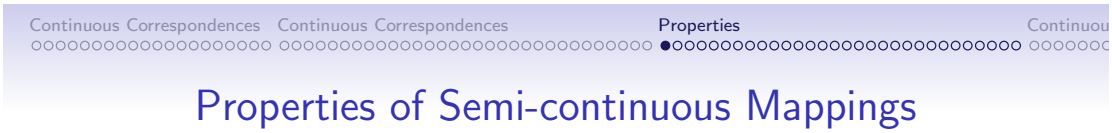
Semi-continuous Mappings

We now prove that $K \subseteq \varphi(K)$. Let a be a point of K; then, for each n, there exists a point x_n in $\varphi^n(S)$ such that $a \in \varphi(x_n)$. The sequence $\{x_n : n \in \mathbb{N}\}$ admits a cluster point x_0 and a sub-sequence $\{x_{k_n} : n \in \mathbb{N}\}$ converges to x_0. Since at most $n-1$ points of this sub-sequence are outside $\varphi^n(S)$ and $\varphi^n(S)$ is compact, we have $x_0 \in \varphi^n(S)$ and so $x_0 \in K$. Also $(x_{k_n}, a) \longrightarrow (x_0, a)$ and $(x_{k_n}, a) \in G(\varphi) = \sum_{x \in S} \varphi(x)$ and so, since φ is a closed mapping, $a \in \varphi(x_0)$; therefore $a \in \varphi(K)$. It follows that $K \subseteq \varphi(K)$ and hence that $\varphi(K) = K$. $\square$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“紧空间上的不变紧集”。完整定义、条件和论证见本讲原页第 52 页下方。

54 原课件第 54 页：紧空间上的不变紧集



In this section we compare the properties of lower semi-continuity with those of upper semi-continuity.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“紧空间上的不变紧集”。完整定义、条件和论证见本讲原页第 52 页下方。

55 原课件第 55 页：复合与下半连续并

Properties of Semi-continuous Mappings

Theorem 2.1. If φ_1 is a lower semi-continuous (l.s.c.) correspondence of S into T and φ_2 is a lower semi-continuous (l.s.c.) correspondence of T into Z, the composition product $\varphi = \varphi_2 \cdot \varphi_1$ is a lower semi-continuous (l.s.c.) correspondence of S into Z.

Proof 1. If G is an open set in Z, then

$$\begin{aligned}
\varphi^-(G) &= \{x \in S \mid \varphi(x) \cap G \neq \varnothing\} \\
&= \{x \in S \mid (\varphi_2 \cdot \varphi_1)(x) \cap G \neq \varnothing\} \\
&= \{x \in S \mid \varphi_2(\varphi_1(x)) \cap G \neq \varnothing\} \\
&= \left\{x \in S \mid \varphi_1(x) \cap \varphi_2^-(G) \neq \varnothing\right\} \\
&= \varphi_1^-\left(\varphi_2^-(G)\right).
\end{aligned}$$

Therefore, if φ_1 and φ_2 are lower semi-continuous (l.s.c.), $\varphi^-(G)$ is open and so φ is lower semi-continuous (l.s.c.) □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

复合的下逆像满足 $(\Psi \circ \Phi)^-(V) = \Phi^-(\Psi^-(V))$，故两次下半连续使开集逐次变开；上逆像的全称量词也能逐层写出，得上半连续复合。任意并的下逆像为各下逆像之并，因此任意并保下半连续。每个等式都来自“存在中间选择”或“所有中间选择”，而不是把对应当单值函数计算。

56 原课件第 56 页：复合与下半连续并

Properties of Semi-continuous Mappings

Theorem 2.1. If φ_1 is a lower semi-continuous (l.s.c.) correspondence of S into T and φ_2 is a lower semi-continuous (l.s.c.) correspondence of T into Z, then the composition product $\varphi = \varphi_2 \cdot \varphi_1$ is a lower semi-continuous (l.s.c.) correspondence of S into Z.

Proof 2. If F is a closed set in Z, then

$$\begin{aligned}
\varphi^+(F) &= \{x \in S \mid \varphi(x) \subseteq F\} \\
&= \{x \in S \mid (\varphi_2 \cdot \varphi_1)(x) \subseteq F\} \\
&= \{x \in S \mid \varphi_2(\varphi_1(x)) \subseteq F\} \\
&= \left\{x \in S \mid \varphi_1(x) \subseteq \varphi_2^+(F)\right\} \\
&= \varphi_1^+\left(\varphi_2^+(F)\right).
\end{aligned}$$

Therefore, if φ_1 and φ_2 are lower semi-continuous (l.s.c.), $\varphi^+(F)$ is closed and so φ is lower semi-continuous (l.s.c.) □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“复合与下半连续并”。完整定义、条件和论证见本讲原页第 55 页下方。

57 原课件第 57 页：复合与下半连续并

Properties of Semi-continuous Mappings

Theorem 2.1′. If φ_1 is a upper semi-continuous (u.s.c.) correspondence of S into T and φ_2 is a upper semi-continuous (u.s.c.) correspondence of T into Z, then the composition product $\varphi = \varphi_2 \cdot \varphi_1$ is a upper semi-continuous (u.s.c.) correspondence of S into Z.

Proof 1. If G is an open set in Z, then

$$\begin{aligned}
\varphi^+(G) &= \{x \in S \mid \varphi(x) \subseteq G\} \\
&= \{x \in S \mid (\varphi_2 \cdot \varphi_1)(x) \subseteq G\} \\
&= \{x \in S \mid \varphi_2(\varphi_1(x)) \subseteq G\} \\
&= \left\{x \in S \mid \varphi_1(x) \subseteq \varphi_2^+(G)\right\} \\
&= \varphi_1^+\left(\varphi_2^+(G)\right).
\end{aligned}$$

Therefore, if φ_1 and φ_2 are upper semi-continuous (u.s.c.), $\varphi^+(G)$ is open and so φ is upper semi-continuous (u.s.c.) □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“复合与下半连续并”。完整定义、条件和论证见本讲原页第 55 页下方。

58 原课件第 58 页：复合与下半连续并

Properties of Semi-continuous Mappings

Theorem 2.1′. If φ_1 is a upper semi-continuous (u.s.c.) correspondence of S into T and φ_2 is a upper semi-continuous (u.s.c.) correspondence of T into Z, then the composition product $\varphi = \varphi_2 \cdot \varphi_1$ is a upper semi-continuous (u.s.c.) correspondence of S into Z.

Proof 2. If F is a closed set of Z, then

$$\begin{aligned}
\varphi^{-}(F) &= \{x \in S \mid \varphi(x) \cap F \neq \varnothing\} \\
&= \{x \in S \mid (\varphi_2 \cdot \varphi_1)(x) \cap F \neq \varnothing\} \\
&= \{x \in S \mid \varphi_2(\varphi_1(x)) \cap F \neq \varnothing\} \\
&= \left\{x \in S \mid \varphi_1(x) \subseteq \varphi_2^{-}(F) \neq \varnothing\right\} \\
&= \varphi_1^{-}\left(\varphi_2^{-}(F)\right).
\end{aligned}$$

Therefore if φ_1 and φ_2 are upper semi-continuous (u.s.c.), $\varphi^{-}(F)$ is closed and so φ is upper semi-continuous (u.s.c.) □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“复合与下半连续并”。完整定义、条件和论证见本讲原页第 55 页下方。

59 原课件第 59 页：复合与下半连续并

Properties of Semi-continuous Mappings

Theorem 2.2. If every mapping φ_λ of S into T is a lower semi-continuous (l.s.c.) correspondence for $\lambda \in \Lambda$, then the union $\varphi = \bigcup_{\lambda\in\Lambda} \varphi_\lambda$ is also a lower semi-continuous (l.s.c.) correspondence of S into T.

Proof 1. If G is an open set of T, we have

$$\begin{aligned}
\varphi^-(G) &= \{x \in S \mid \varphi(x) \cap G \neq \varnothing\} \\
&= \left\{x \in S \mid \bigcup_{\lambda\in\Lambda} [\varphi_\lambda(x) \cap G] \neq \varnothing\right\} \\
&= \bigcup_{\lambda\in\Lambda} \{x \in S \mid \varphi_\lambda(x) \cap G \neq \varnothing\} \\
&= \bigcup_{\lambda\in\Lambda} \varphi_\lambda^-(G).
\end{aligned}$$

Therefore the set $\varphi^-(G)$ is open and so φ is l.s.c. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“复合与下半连续并”。完整定义、条件和论证见本讲原页第 55 页下方。

60 原课件第 60 页：复合与下半连续并

Properties of Semi-continuous Mappings

Theorem 2.2. If every mapping φ_λ of S into T is a lower semi-continuous (l.s.c.) correspondence for $\lambda \in \Lambda$, then the union $\varphi = \bigcup_{\lambda\in\Lambda} \varphi_\lambda$ is also a lower semi-continuous (l.s.c.) correspondence of S into T.

Proof 2. If F is a closed set of T, we have

$$\begin{aligned}
\varphi^+(F) &= \{x \in S \mid \varphi(x) \subseteq F\} \\
&= \left\{x \in S \mid \bigcup_{\lambda\in\Lambda} \varphi_\lambda(x) \subseteq F\right\} \\
&= \left\{x \in S \mid \mathbf{C}F \subseteq \bigcap_{\lambda\in\Lambda} \mathbf{C}(\varphi_\lambda(x))\right\} \\
&= \bigcap_{\lambda\in\Lambda} \{x \in S \mid \mathbf{C}F \subseteq \mathbf{C}(\varphi_\lambda(x))\} \\
&= \bigcap_{\lambda\in\Lambda} \{x \in S \mid \varphi_\lambda(x) \subseteq F\} \\
&= \bigcap_{\lambda\in\Lambda} \varphi_\lambda^+(F).
\end{aligned}$$

Therefore the set $\varphi^+(F)$ is closed and so φ is l.s.c. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“复合与下半连续并”。完整定义、条件和论证见本讲原页第 55 页下方。

61 原课件第 61 页：对应交集的条件与伪定理

Properties of Semi-continuous Mappings

Theorem 2.2′. If every mapping φ_λ of S into T is a compact-valued u.s.c. correspondence for $\lambda \in \Lambda$, then the intersection $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ is also a compact-valued u.s.c. correspondence of S into T.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

紧值上半连续对应的任意交是闭图，固定一个紧值上半连续因子作容器，运用第45页交集定理得紧值上半连续（交值可能为空）。第63–64页删掉紧值后的证明不能沿用，尤其 $(\bigcap_i \Phi_i)^-(F)$ 一般不等于 $\bigcap_i \Phi_i^-(F)$：各因子与 F 相交可能在不同点。第65–66页明确为伪定理：下半连续的有限交不必下半连续。反例 $\Phi_1(x) = \{0\}$，$\Phi_2(x) = \{x\}$，交仅在0非空；若要求非空值，可改 $\Phi_1(x) = \{0, 1\}, \Phi_2(x) = \{x, 1\}$，在0交含0与1，非零附近只含1，不下半连续。

62 原课件第 62 页：对应交集的条件与伪定理

Properties of Semi-continuous Mappings

Proof. By Theorem 1.5, the mapping φ_λ is compact-valued u.s.c., then the mapping $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ is closed. Suppose that $y_0 \notin \varphi(x_0) = \bigcap_{\lambda\in\Lambda} \varphi_\lambda(x_0)$, then there exists $\lambda_0 \in$ such that $y_0 \notin \varphi_{\lambda_0}(x_0)$. There exist neighbourhood $U(x_0)$ and neighbourhood $V(y_0)$ such that

$$x \in U(x_0) \qquad \Longrightarrow \qquad \varphi_{\lambda_0}(x) \cap V(y_0) = \varnothing,$$

that is, $\varphi_{\lambda_0}(U(x_0)) \cap V(y_0) = \varnothing$, then $\bigcap_{\lambda\in\Lambda} \varphi_\lambda(U(x_0)) \cap V(y_0) = \varnothing$ and hence $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ is closed.

The intersection $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ of a family of compact-valued mappings φ_λ is compact-valued from Non-empty Finite-intersection Theorem.

On the other hand, $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda = \varphi_{\gamma'} \cap \bigcap_{\gamma\neq\gamma'} \varphi_\lambda$. The mapping $\varphi_{\gamma'}$ is compact-valued u.s.c., and the mapping $\bigcap_{\gamma\neq\gamma'} \varphi_\lambda$ is closed, therefore, by Theorem 1.7, the mapping φ is u.s.c. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“对应交集的条件与伪定理”。完整定义、条件和论证见本讲原页第 61 页下方。

63 原课件第 63 页：对应交集的条件与伪定理

Properties of Semi-continuous Mappings

Theorem 2.2′. If every mapping φ_λ of S into T is a upper semi-continuous (u.s.c.) correspondence for $\lambda \in \Lambda$, then the intersection $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ is also a upper semi-continuous (u.s.c.) correspondence of S into T.

Proof 1. If G is an open set of T, we have

$$\begin{aligned}
\varphi^+(G) &= \{x \in S \mid \varphi(x) \subseteq G\} = \left\{x \in S \mid \bigcap_{\lambda\in\Lambda} \varphi_\lambda(x) \subseteq G\right\} \\
&= \left\{x \in S \mid \mathbf{C}_T G \subseteq \mathbf{C}_T\left(\bigcap_{\lambda\in\Lambda} \varphi_\lambda(x)\right)\right\} = \left\{x \in S \mid \mathbf{C}_T G \subseteq \bigcup_{\lambda\in\Lambda} \mathbf{C}_T\left(\varphi_\lambda(x)\right)\right\} \\
&= \bigcup_{\lambda\in\Lambda} \{x \in S \mid \mathbf{C}_T G \subseteq \mathbf{C}_T\left(\varphi_\lambda(x)\right)\} = \bigcup_{\lambda\in\Lambda} \{x \in S \mid \varphi_\lambda(x) \subseteq G\} = \bigcup_{\lambda\in\Lambda} \varphi_\lambda^+(G)
\end{aligned}$$

Therefore the set $\varphi^+(G)$ is open and so φ is u.s.c. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“对应交集的条件与伪定理”。完整定义、条件和论证见本讲原页第 61 页下方。

64 原课件第 64 页：对应交集的条件与伪定理

Properties of Semi-continuous Mappings

Theorem 2.2′. If every mapping φ_λ of S into T is a upper semi-continuous (u.s.c.) correspondence for $\lambda \in \Lambda$, then the intersection $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ is also a upper semi-continuous (u.s.c.) correspondence of S into T.

Proof 2. If F is a closed set of T, we have

$$
\begin{aligned}
\varphi^-(F) &= \{x \in S \mid \varphi(x) \cap F \neq \varnothing\} = \left\{x \in S \mid \bigcap_{\lambda\in\Lambda} \varphi_\lambda(x) \cap F \neq \varnothing\right\} \\
&= \mathbf{C}_S\left\{x \in S \mid \bigcap_{\lambda\in\Lambda} \varphi_\lambda(x) \cap F = \varnothing\right\} = \mathbf{C}_S\left\{x \in S \mid F \subseteq \mathbf{C}_T\left(\bigcap_{\lambda\in\Lambda} \varphi_\lambda(x)\right)\right\} \\
&= \mathbf{C}_S\left\{x \in S \mid F \subseteq \bigcup_{\lambda\in\Lambda} \mathbf{C}_T\left(\varphi_\lambda(x)\right)\right\} = \mathbf{C}_S\left(\bigcup_{\lambda\in\Lambda} \{x \in S \mid F \subseteq \mathbf{C}_T\left(\varphi_\lambda(x)\right)\}\right) \\
&= \mathbf{C}_S\left(\bigcup_{\lambda\in\Lambda} \{x \in S \mid \varphi_\lambda(x) \subseteq \mathbf{C}_T F\}\right) = \mathbf{C}_S\left(\bigcup_{\lambda\in\Lambda} \{x \in S \mid \varphi_\lambda(x) \cap F = \varnothing\}\right) \\
&= \bigcap_{\lambda\in\Lambda} \{x \in S \mid \varphi_\lambda(x) \cap F \neq \varnothing\} = \bigcap_{\lambda\in\Lambda} \varphi_\lambda^-(F).
\end{aligned}
$$

Therefore the set $\varphi^-(F)$ is closed and so φ is u.s.c. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“对应交集的条件与伪定理”。完整定义、条件和论证见本讲原页第 61 页下方。

65 原课件第 65 页：对应交集的条件与伪定理

Properties of Semi-continuous Mappings

Pseudo-Theorem 2.3. If every mapping φ_n of S into T is a lower semi-continuous (l.s.c.) correspondence for $n = 1, \cdots, N$, then the intersection $\varphi = \bigcap_{n=1}^{N} \varphi_n$ is also a lower semi-continuous (l.s.c.) correspondence of S into T.

Proof 1. If G is an open set in T, we have

$$
\begin{aligned}
\varphi^{-}(G) &= \{x \in S \mid \varphi(x) \cap G \neq \varnothing\} = \left\{x \in S \mid \bigcap_{n=1}^{N} \varphi_n(x) \cap G \neq \varnothing\right\} \\
&= \mathbf{C}_S \left\{x \in S \mid \bigcap_{n=1}^{N} \varphi_n(x) \cap G = \varnothing\right\} = \mathbf{C}_S \left\{x \in S \mid \bigcap_{n=1}^{N} \varphi_n(x) \subseteq \mathbf{C}_T G\right\} \\
&= \mathbf{C}_S \left\{x \in S \mid G \subseteq \bigcup_{n=1}^{N} \mathbf{C}_T\left(\varphi_n(x)\right)\right\} = \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid G \subseteq \mathbf{C}_T\left(\varphi_n(x)\right)\}\right) \\
&= \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \subseteq \mathbf{C}_T G\}\right) = \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \cap G = \varnothing\}\right) \\
&= \bigcap_{n=1}^{N} \{x \in S \mid \varphi_n(x) \cap G \neq \varnothing\} = \bigcap_{n=1}^{N} \varphi_n^{-}(G).
\end{aligned}
$$

Therefore $\varphi^{-}(G)$ is an open set and so φ is l.s.c. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“对应交集的条件与伪定理”。完整定义、条件和论证见本讲原页第 61 页下方。

66 原课件第 66 页：对应交集的条件与伪定理

Properties of Semi-continuous Mappings

Pseudo-Theorem 2.3. If every mapping φ_n of S into T is a lower semi-continuous (l.s.c.) correspondence for $n = 1, \cdots, N$, then the intersection $\varphi = \bigcap_{n=1}^{N} \varphi_n$ is also a lower semi-continuous (l.s.c.) correspondence of S into T.

Proof 2. If F is a closed set in T, we have

$$\begin{aligned}
\varphi^+(F) &= \{x \in S \mid \varphi(x) \subseteq F\} = \left\{x \in S \mid \bigcap_{n=1}^{N} \varphi_n(x) \subseteq F\right\} \\
&= \left\{x \in S \mid \mathbf{C}_T F \subseteq \mathbf{C}_T \left(\bigcap_{n=1}^{N} \varphi_n(x)\right)\right\} \\
&= \left\{x \in S \mid \mathbf{C}_T F \subseteq \bigcup_{n=1}^{N} \mathbf{C}_T\left(\varphi_n(x)\right)\right\} \\
&= \bigcup_{n=1}^{N} \{x \in S \mid \mathbf{C}_T F \subseteq \mathbf{C}_T\left(\varphi_n(x)\right)\} \\
&= \bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \subseteq F\} = \bigcup_{n=1}^{N} \varphi_n^+(F).
\end{aligned}$$

Therefore $\varphi^+(F)$ is a closed set and so φ is l.s.c. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“对应交集的条件与伪定理”。完整定义、条件和论证见本讲原页第 61 页下方。

67 原课件第 67 页：有限并保持上半连续

Properties of Semi-continuous Mappings

Theorem 2.3′. If every mapping φ_n of S into T is a compacted-valued u.s.c. correspondence for $n = 1, \cdots, N$, then the union $\varphi = \bigcup_{n=1}^{N} \varphi_n$ is also a compacted-valued u.s.c. correspondence of S into T.

Proof 1. The set $\varphi(x) = \bigcup_{n=1}^{N} \varphi_n(x)$ is compact for all x. Moreover, if G is an open set in T, we have

$$\begin{aligned}
\varphi^+(G) &= \{x \in S \mid \varphi(x) \subseteq G\} = \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \subseteq G\right\} \\
&= \left\{x \in S \mid \mathbf{C}_T G \subseteq \mathbf{C}_T \left(\bigcup_{n=1}^{N} \varphi_n(x)\right)\right\} = \left\{x \in S \mid \mathbf{C}_T G \subseteq \bigcap_{n=1}^{N} \mathbf{C}_T\left(\varphi_n(x)\right)\right\} \\
&= \bigcap_{n=1}^{N} \{x \in S \mid \mathbf{C}_T G \subseteq \mathbf{C}_T\left(\varphi_n(x)\right)\} = \bigcap_{n=1}^{N} \{x \in S \mid \varphi_n(x) \subseteq G\} = \bigcap_{n=1}^{N} \varphi_n^+(G).
\end{aligned}$$

Therefore $\varphi^+(G)$ is an open set and so φ is u.s.c. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

若开集 G 包含 $\bigcup_{i=1}^{m} \Phi_i(x)$，它包含每个 $\Phi_i(x)$。为每个因子取邻域 U_i 使其值在 G 内，有限交 U 仍为邻域，故并值在 G 内。紧值时有限并亦紧。原页逆像公式须注意：并对应的上逆像是各上逆像的交，下逆像是各下逆像的并。不能把两者写成同一种交并。

68 原课件第 68 页：有限并保持上半连续

Properties of Semi-continuous Mappings

Theorem 2.3′. If every mapping φ_n of S into T is a compacted-valued u.s.c. correspondence for $n = 1, \cdots, N$, then the union $\varphi = \bigcup_{n=1}^{N} \varphi_n$ is also a compacted-valued u.s.c. correspondence of S into T.

Proof 2. The set $\varphi(x) = \bigcup_{n=1}^{N} \varphi_n(x)$ is compact for all x. Moreover, if F is a closed set in T, we have

$$
\begin{aligned}
\varphi^{-}(F) &= \{x \in S \mid \varphi(x) \cap F \neq \varnothing\} = \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \cap F \neq \varnothing\right\} \\
&= \mathbf{C}_S \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \cap F = \varnothing\right\} = \mathbf{C}_S \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \subseteq \mathbf{C}_T F\right\} \\
&= \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \subseteq \mathbf{C}_T F\}\right) = \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \cap F \neq \varnothing\}\right) \\
&= \bigcap_{n=1}^{N} \{x \in S \mid \varphi_n(x) \cap F = \varnothing\} = \bigcap_{n=1}^{N} \varphi_n^{-}(F).
\end{aligned}
$$

Therefore $\varphi^{-}(F)$ is a closed set and so φ is u.s.c. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“有限并保持上半连续”。完整定义、条件和论证见本讲原页第 67 页下方。

69 原课件第 69 页：有限并保持上半连续

Properties of Semi-continuous Mappings

Theorem 2.3′. If every mapping φ_n of S into T is a upper semi-continuous (u.s.c.) correspondence for $n = 1, \cdots, N$, then the union $\varphi = \bigcup_{n=1}^{N} \varphi_n$ is also a upper semi-continuous (u.s.c.) correspondence of S into T.

Proof 1. If G is an open set in T, we have

$$\begin{aligned}
\varphi^+(G) &= \{x \in S \mid \varphi(x) \subseteq G\} = \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \subseteq G\right\} \\
&= \left\{x \in S \mid \mathbf{C}_T G \subseteq \mathbf{C}_T \left(\bigcup_{n=1}^{N} \varphi_n(x)\right)\right\} \\
&= \left\{x \in S \mid \mathbf{C}_T G \subseteq \bigcap_{n=1}^{N} \mathbf{C}_T\left(\varphi_n(x)\right)\right\} \\
&= \bigcap_{n=1}^{N} \{x \in S \mid \mathbf{C}_T G \subseteq \mathbf{C}_T\left(\varphi_n(x)\right)\} \\
&= \bigcap_{n=1}^{N} \{x \in S \mid \varphi_n(x) \subseteq G\} = \bigcap_{n=1}^{N} \varphi_n^+(G).
\end{aligned}$$

Therefore $\varphi^+(G)$ is an open set and so φ is u.s.c. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“有限并保持上半连续”。完整定义、条件和论证见本讲原页第 67 页下方。

70 原课件第 70 页：有限并保持上半连续

Properties of Semi-continuous Mappings

Theorem 2.3′. If every mapping φ_n of S into T is a upper semi-continuous (u.s.c.) correspondence for $n = 1, \cdots, N$, then the union $\varphi = \bigcup_{n=1}^{N} \varphi_n$ is also a upper semi-continuous (u.s.c.) correspondence of S into T.

Proof 2. If F is a closed set in T, we have

$$\begin{aligned}
\varphi^{-}(F) &= \{x \in S \mid \varphi(x) \cap F \neq \varnothing\} = \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \cap F \neq \varnothing\right\} \\
&= \mathbf{C}_S \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \cap F = \varnothing\right\} = \mathbf{C}_S \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \subseteq \mathbf{C}_T F\right\} \\
&= \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \subseteq \mathbf{C}_T F\}\right) = \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \cap F \neq \varnothing\}\right) \\
&= \bigcap_{n=1}^{N} \{x \in S \mid \varphi_n(x) \cap F = \varnothing\} = \bigcap_{n=1}^{N} \varphi_n^{-}(F).
\end{aligned}$$

Therefore $\varphi^{-}(F)$ is a closed set and so φ is u.s.c. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“有限并保持上半连续”。完整定义、条件和论证见本讲原页第 67 页下方。

71 原课件第 71 页：有限积对应的半连续性

Properties of Semi-continuous Mappings

Theorem 2.4. If every mapping φ_n is a lower semi-continuous (l.s.c.) correspondence of S into T_n for $n = 1, \cdots, N$, then the Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ is also a lower semi-continuous (l.s.c.) correspondence of S into $T = \prod_{n=1}^{N} T_n$.

Proof 1. Let $G = \bigcup_k E^k$ be the union of the elementary open sets E^k in T. Since $E^k = \prod_{n=1}^{N} G_n^k$, where G_n^k is an open set in T_n, we have

$$\begin{aligned}\varphi^-\left(E^k\right) &= \left\{x \in S \mid \varphi(x) \cap E^k \neq \varnothing\right\} = \left\{x \in S \mid \prod_{n=1}^{N} \varphi_n(x) \cap \prod_{n=1}^{N} G_n^k \neq \varnothing\right\} \\ &= \left\{x \in S \mid \prod_{n=1}^{N} \left(\varphi_n(x) \cap G_n^k\right) \neq \varnothing\right\} = \bigcap_{n=1}^{N} \left\{x \in S \mid \varphi_n(x) \cap G_n^k \neq \varnothing\right\} = \bigcap_{n=1}^{N} \varphi_n^-\left(G_n^k\right).\end{aligned}$$

Then $\varphi^-\left(E^k\right)$ is an open set and therefore so is $\varphi^-(G) = \bigcup_k \varphi^-\left(E^k\right)$.

$$\begin{aligned}\varphi^-(G) &= \varphi^-\left(\bigcup_k E^k\right) = \left\{x \in S \mid \varphi(x) \cap \left(\bigcup_k E^k\right) \neq \varnothing\right\} \\ &= \left\{x \in S \mid \bigcup_k \left[\varphi(x) \cap E^k\right] \neq \varnothing\right\} \\ &= \bigcup_k \left\{x \in S \mid \varphi(x) \cap E^k \neq \varnothing\right\} = \bigcup_k \varphi^-\left(E^k\right).\end{aligned}$$

Hence φ is l.s.c. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

下半连续积：给定值点 $y = (y_i)$ 和含它的开集，取乘积开邻域 $\prod_i V_i$ 包含于其中。各因子的下半连续保证附近可分别选 $y_i' \in \Phi_i(x') \cap V_i$，于是组成积中选择。紧值上半连续积：给任意输入收敛序列及积值选择，依次抽每个因子的收敛子列，有限次后所有坐标同时收敛且极限在积值中；若不满足上半连续，选逃离开邻域的序列即矛盾。没有紧值的第73–74页不能用上述紧性证明，也不能把任意开集当单个开矩形；本讲使用有紧值条件的正确版本。

72 原课件第 72 页：有限积对应的半连续性

Properties of Semi-continuous Mappings

Theorem 2.4. If every mapping φ_n is a lower semi-continuous (l.s.c.) correspondence of S into T_n for $n = 1, \cdots, N$, then the Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ is also a lower semi-continuous (l.s.c.) correspondence of S into $T = \prod_{n=1}^{N} T_n$.

Proof 2. Let $F = \bigcap_k E^k$ be the intersection of the elementary closed sets E^k in T. Since $E^k = \prod_{n=1}^{N} F_n^k$, where F_n^k is a closed set in T_n, we have

$$\begin{aligned} \varphi^+\left(E^k\right) &= \left\{x \in S \mid \varphi(x) \subseteq E^k\right\} = \left\{x \in S \mid \prod_{n=1}^{N} \varphi_n(x) \subseteq \prod_{n=1}^{N} F_n^k\right\} \\ &= \bigcap_{n=1}^{N} \left\{x \in S \mid \varphi_n(x) \subseteq F_n^k\right\} = \bigcap_{n=1}^{N} \varphi_n^+\left(F_n^k\right). \end{aligned}$$

Then $\varphi^+\left(E^k\right)$ is a closed set and therefore so is $\varphi^+(F) = \bigcap_k \varphi^+\left(E^k\right)$.

$$\begin{aligned} \varphi^+(F) &= \varphi^+\left(\bigcap_k E^k\right) = \left\{x \in S \mid \varphi(x) \subseteq \bigcap_k E^k\right\} \\ &= \bigcap_k \left\{x \in S \mid \varphi(x) \subseteq E^k\right\} = \bigcap_k \varphi^+\left(E^k\right). \end{aligned}$$

Hence φ is l.s.c. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“有限积对应的半连续性”。完整定义、条件和论证见本讲原页第 71 页下方。

73 原课件第 73 页：有限积对应的半连续性

Properties of Semi-continuous Mappings

Theorem 2.4′. If every mapping φ_n is a upper semi-continuous (u.s.c.) correspondence of S into T_n for $n = 1, \cdots, N$, then the Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ is also a upper semi-continuous (u.s.c.) correspondence of S into $T = \prod_{n=1}^{N} Y_n$.

Proof 1. Let $G = \bigcup_k E^k$ be the union of the elementary open sets E^k in T. Since $E^k = \prod_{n=1}^{N} G_n^k$, where G_n^k is an open set in T_n, we have

$$\begin{aligned} \varphi^+\left(E^k\right) &= \left\{x \in S \mid \varphi(x) \subseteq E^k\right\} = \left\{x \in S \mid \prod_{n=1}^{N} \varphi_n(x) \subseteq \prod_{n=1}^{N} G_n^k\right\} \\ &= \bigcap_{n=1}^{N} \left\{x \in S \mid \varphi_n(x) \subseteq G_n^k\right\} = \bigcap_{n=1}^{N} \varphi_n^+\left(G_n^k\right). \end{aligned}$$

Then $\varphi^+\left(E^k\right)$ is an open set and therefore so is $\varphi^+(G) = \bigcup_k \varphi^+\left(E^k\right)$.

$$\begin{aligned} \varphi^+(G) &= \varphi^+\left(\bigcup_k E^k\right) = \left\{x \in S \mid \varphi(x) \subseteq \bigcup_k E^k\right\} \\ &= \bigcup_k \left\{x \in S \mid \varphi(x) \subseteq E^k\right\} = \bigcup_k \varphi^+\left(E^k\right). \end{aligned}$$

Hence φ is u.s.c. □

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本页继续“有限积对应的半连续性”。完整定义、条件和论证见本讲原页第 71 页下方。

74 原课件第 74 页：有限积对应的半连续性

Properties of Semi-continuous Mappings

Theorem 2.4′. If every mapping φ_n is a upper semi-continuous (u.s.c.) correspondence of S into T_n for $n = 1, \cdots, N$, then the Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ is also a upper semi-continuous (u.s.c.) correspondence of S into $T = \prod_{n=1}^{N} T_n$.

Proof 2. Let $F = \bigcap_k E^k$ be the intersection of the elementary closed sets E^k in T. Since $E^k = \prod_{n=1}^{N} F_n^k$, where F_n^k is a closed set in T_n, we have

$$\begin{aligned}\varphi^- \left(E^k\right) &= \left\{x \in S \mid \varphi(x) \cap E^k \neq \varnothing\right\} = \left\{x \in S \mid \prod_{n=1}^{N} \varphi_n(x) \cap \prod_{n=1}^{N} F_n^k \neq \varnothing\right\} \\ &= \left\{x \in S \mid \prod_{n=1}^{N} \left(\varphi_n(x) \cap F_n^k\right) \neq \varnothing\right\} = \bigcap_{n=1}^{N} \left\{x \in S \mid \varphi_n(x) \cap F_n^k \neq \varnothing\right\} = \bigcap_{n=1}^{N} \varphi_n^- \left(F_n^k\right).\end{aligned}$$

Then $\varphi^- \left(E^k\right)$ is a closed set and therefore so is $\varphi^-(F) = \bigcap_k \varphi^- \left(E^k\right)$.

$$\begin{aligned}\varphi^-(F) &= \varphi^- \left(\bigcap_k E^k\right) = \left\{x \in S \mid \varphi(x) \cap \left(\bigcap_k E^k\right) \neq \varnothing\right\} \\ &= \left\{x \in S \mid \bigcap_k \left[\varphi(x) \cap E^k\right] \neq \varnothing\right\} \\ &= \bigcap_k \left\{x \in S \mid \varphi(x) \cap E^k \neq \varnothing\right\} = \bigcap_k \varphi^- \left(E^k\right).\end{aligned}$$

Hence φ is u.s.c. □

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本页继续“有限积对应的半连续性”。完整定义、条件和论证见本讲原页第 71 页下方。

75 原课件第 75 页：有限积对应的半连续性

Properties of Semi-continuous Mappings

Theorem 2.4′. If every mapping φ_n is a compact-valued u.s.c. correspondence of S into T_n for $n = 1, \cdots, N$, then the Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ is also a compact-valued u.s.c. correspondence of S into $T = \prod_{n=1}^{N} T_n$.

Proof. To fix our ideas, we consider the case $N = 2$. Since $\varphi_1(x)$ and $\varphi_2(x)$ are compact, so is $\varphi(x) = \varphi_1(x) \times \varphi_2(x)$, by Tychonoff's Theorem.

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76 原课件第 76 页：有限积对应的半连续性

Properties of Semi-continuous Mappings

Let G be an open set in T and. let a be an element of $\varphi^+(G)$. Since $\varphi(a)$ is compact and is contained in G, it can be covered by a finite number of elementary open sets contained in G; let $E^1, \cdots, E^K$ be such a finite covering. Suppose that $(y_1, y_2) \in \varphi_1(a) \times \varphi_2(a)$; let $E(y_1)$ denote the union of the E^k meeting the set $\{y_1\} \times \varphi_2(a)$ and let $E(y_2)$ be the union of the E^k meeting the set $\varphi_1(a) \times \{y_2\}$. The projection $\pi_2 E(y_1)$ of the set $E(y_1)$ on the space Y_2 is open, by Theorem 2 of §9, Chapter IV. As y_1 varies the sets. $\pi_2 E(y_1)$ are finite in number and therefore their intersection is open; similarly the intersection of the sets $\pi_1 E(y_2)$ is open. We write

$$G_1 = \bigcap_{y_2 \in \varphi_2(a)} \pi_1 E(y_2) \qquad \text{and} \qquad G_2 = \bigcap_{y_1 \in \varphi_1(a)} \pi_2 E(y_1).$$

Then $E = G_1 \times G_2$ is an elementary open set of T and

$$\varphi^+(E) = \{x \mid \varphi_1(x) \times \varphi_2(x) \subseteq G_1 \times G_2\} = \left[\varphi_1^+(G_1) \cap \varphi_2^+(G_2)\right] \cup \varphi_1^+(\varnothing) \cup \varphi_2^+(\varnothing).$$

Therefore $\varphi^+(E)$ is an open set of S; since $\varphi(a) \subseteq E \subseteq G$, we also have

$$a \in \varphi^+(E) \subseteq \varphi^+(G)$$

Therefore the set $\varphi^+(G)$ is a neighbourhood of each of its points and so is open; hence φ is compact-valued u.s.c. □

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本页继续“有限积对应的半连续性”。完整定义、条件和论证见本讲原页第 71 页下方。

77 原课件第 77 页：和对应与凸包

Properties of Semi-continuous Mappings

Theorem 2.5′. Let S and T be topological vector spaces. If every mapping $\varphi_n : X \to Y$ is a compact-valued u.s.c. correspondence for $n = 1, \cdots, N$, then the sum $\varphi = \sum_{n=1}^{N} \varphi_n$ is a compact-valued u.s.c. correspondence.

Theorem 2.5′. The sum $\varphi = \sum_{n=1}^{N} \varphi_n$ of a finite number of compact-valued u.s.c. correspondences φ_n of S into T is a compact-valued u.s.c. correspondence of S into T.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

有限和是积对应经连续加法映射的像，故紧值上半连续性由积与复合得到。下半连续凸包：任取值中一点 $y = \sum_{i=1}^{m} \lambda_i y_i$，对每个 y_i 用下半连续选取随输入趋近的 y_{in}，保持同一组权重，便得凸组合趋于 y。紧值上半连续凸包在 $\mathbb{R}^N$ 也成立：Carathéodory 定理把每点表示为至多 $N+1$ 个点的凸组合，权重在紧单纯形中；对权重和值点抽子列得极限仍在凸包。有限维假设不可遗漏。

78 原课件第 78 页：和对应与凸包

Properties of Semi-continuous Mappings

Proof. We use mathematical induction to verify this theorem.

(1) If $\varphi_n(x_0)$ is compact for $n = 1, \cdots, N$, then $\varphi(x_0) = \sum_{n=1}^{N} \varphi_n(x_0)$ is compact.

(2) For any open set $G \subseteq T$ with $\varphi(x_0) = \sum_{n=1}^{N} \varphi_n(x_0) \subseteq G$, there exists a neighborhood $U(x_0)$ such that $x \in U(x_0)$ implies $\varphi(x) = \sum_{n=1}^{N} \varphi_n(x) \subseteq G$.

(1) Let T be a topological vector space. Define $f : T^N \to T$ by $f(y_1, \cdots, y_N) = \sum_{n=1}^{N} y_n$, then f is a continuous function, $f(\varphi_1 x_0, \cdots, \varphi_n(x_0)) = \sum_{n=1}^{N} \varphi_n(x_0)$. As you know, $\varphi_n(x_0)$ is compact for $n = 1, \cdots, N$, then $\varphi(x_0) = \sum_{n=1}^{N} \varphi_n(x_0)$ is compact.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“和对应与凸包”。完整定义、条件和论证见本讲原页第 77 页下方。

79 原课件第 79 页：和对应与凸包

Properties of Semi-continuous Mappings

(2) For any open set $G \subseteq T$ with $\varphi(x_0) = \sum_{n=1}^{N} \varphi_n(x_0) \subseteq G$, there exists a neighborhood $V = U(0)$ of origin such that $\sum_{n=1}^{N} \varphi_n(x_0) + V \subseteq G$, there exist N neighborhoods $V_1, \cdots, V_N$ with $\sum_{n=1}^{N} V_n \subseteq V$ such that $\sum_{n=1}^{N} [\varphi_n(x_0) + V_n] = \sum_{n=1}^{N} \varphi_n(x_0) + V \subseteq G$. $\varphi_n : X \to Y$ is u.s.c. for $n = 1, \cdots, N$, then $\varphi_n(x_0) \subseteq \varphi_n(x_0) + V_n$, there exists a neighborhood $U_n(x_0)$ such that

$$x \in U_n(x_0) \quad \Longrightarrow \quad x \in \varphi_n(x_0) + V_n.$$

For any $x \in N(x_0) \equiv \bigcap_{n=1}^{N} U_n(x_0)$,

$$\varphi(x) = \sum_{n=1}^{N} \varphi_n(x) \subseteq \sum_{n=1}^{N} [\varphi_n(x_0) + V_n] \subseteq \sum_{n=1}^{N} \varphi_n(x_0) + V \subseteq G.$$

Thus $\varphi = \sum_{n=1}^{N} \varphi_n$ is compact-valued and u.s.c. □

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本页继续“和对应与凸包”。完整定义、条件和论证见本讲原页第 77 页下方。

80 原课件第 80 页：和对应与凸包

Properties of Semi-continuous Mappings

Theorem 2.6. Let T be a linear vector space. If $\varphi : S \to \mathbb{R}^N$ is a lower semi-continuous (l.s.c.) correspondence, then its convex envelope $\mathbf{CE}(\varphi)$ is a lower semi-continuous (l.s.c.) correspondence.

Proof. According to Caratheodory Characterization Theorem, for any $y_0 \in \mathbf{CE}(\varphi)(x_0) = \mathbf{CE}(\varphi(x_0))$,

$$y_0 = \sum_{k=1}^{K} \lambda^k z_0^k, \qquad z_0^k \in \varphi(x_0), \quad \sum_{k=1}^{K} \lambda_n^k = 1, \quad \lambda_n^k \geqslant 0.$$

For any $x_n \to x_0$, there exists a sequence $z_n^k \to z_0^k$ such that $z_n^k \in \varphi(x_n)$, $y_n = \sum_{k=1}^{K} \lambda^k z_n^k \in \varphi(x_0) \subseteq \mathbf{CE}(\varphi(x_n))$. Therefore $\mathbf{CE}(\varphi)$ is lower semi-continuous (l.s.c.). □

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本页继续“和对应与凸包”。完整定义、条件和论证见本讲原页第 77 页下方。

81 原课件第 81 页：和对应与凸包

Properties of Semi-continuous Mappings

Theorem 2.6′. If $\varphi : S \to \mathbb{R}^N$ is a compact-valued u.s.c. correspondence, then its convex envelope $\mathbf{CE}(\varphi)$ is a compact-valued u.s.c. correspondence.

Proof. According to Caratheodory Characterization Theorem, for any $y_n \in \mathbf{CE}(\varphi)(x_n) = \mathbf{CE}(\varphi(x_n))$,

$$y_n = \sum_{k=1}^{K} \lambda_n^k z_n^k, \qquad z_n^k \in \varphi(x_n), \quad \sum_{k=1}^{K} \lambda_n^k = 1, \quad \lambda_n^k \geqslant 0.$$

For any $x_n \to x_0$, there exists a convergent subsequence $\{z_{n'}^k, n' \in \mathbb{N}\} \subseteq \{z_n^k, n \in \mathbb{N}\}$ such that $\lim_{n'} z_{n'}^k \to z_0^k \in \varphi(x_0)$, then there corresponds a subsequence $\{y_{n'}, n' \in \mathbb{N}\} \subseteq \{y_n, n \in \mathbb{N}\}$ such that $y_{n'} = \sum_{k=1}^{K} \lambda_{n'}^k z_{n'}^k \to \sum_{k=1}^{K} \lambda_n^k z_0^k = y_0 \in \varphi(x_0) \subseteq \mathbf{CE}(\varphi(x_0))$. Therefore $\mathbf{CE}(\varphi)$ is compact-valued and u.s.c. □

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82 原课件第 82 页：闭包对应与内半连续

Properties of Semi-continuous Mappings

Theorem 2.7. $\varphi : S \to T$ is a lower semi-continuous (l.s.c.) correspondence if and only if its convex hull $\mathbf{CH}(\varphi) = \overline{\varphi}$ is a lower semi-continuous (l.s.c.) correspondence.

Proof. For any open set $G \subseteq T$ with $\varphi(x_0) \subseteq G$ equivalently $\overline{\varphi}(x_0) \subseteq G$, there exists a neighbourhood $U(x_0)$ such that, for any $x \in U(x_0)$, $\varphi(x) \subseteq G$ equivalently $\overline{\varphi}(x) \subseteq G$. Thus φ is l.s.c. if and only if its convex hull $\mathbf{CH}(\varphi) = \overline{\varphi}$ is l.s.c. □

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第82–83页的横线运算是闭包，不应标为 convex hull。Φ 与逐点闭包的下逆像对开集相同，因为开集遇到闭包就遇到原集合，故下半连续等价。上半连续的闭包版本用正则性：给开集 G 包含闭包值，先取开邻域 V 使 $\Phi(x) \subset V \subset \overline{V} \subset G$（紧值时可有限选取），再用上半连续。度量空间中下半连续等价于：任意 $x_n \to x, y \in \Phi(x)$ 可选 $y_n \in \Phi(x_n)$ 趋于 y。证明依次对球 $B(y, 1/k)$ 取输入邻域，再用足够大的指标分段选择。

83 原课件第 83 页：闭包对应与内半连续

Properties of Semi-continuous Mappings

Theorem 2.7′**.** Let T be a regular metric space. If $\varphi : S \to T$ is a u.s.c. correspondence, then its convex hull $\mathbf{CH}(\varphi) = \overline{\varphi}$ is a u.s.c. correspondence.

Proof. For any open set $G \subseteq T$ with $\overline{\varphi}(x_0) \subseteq G$, $\overline{\varphi}(x_0) \cap (\mathbf{C}_T G) = \varnothing$. According to the regularity of metric space, there exists an open set G' and O such that

$$\overline{\varphi}(x_0) \subseteq G', \qquad \mathbf{C}_T G \subseteq O, \qquad G' \cap O = \varnothing$$

As φ is u.s.c., there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \subseteq G',$$

that is, $\varphi(U(x_0)) \subseteq G'$. Since $G' \cap (\mathbf{C}_T G) = \varnothing$, then $\overline{\varphi}(U(x_0)) \subseteq G$. Therefore $\overline{\varphi}$ is u.s.c. □

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本页继续“闭包对应与内半连续”。完整定义、条件和论证见本讲原页第 82 页下方。

84 原课件第 84 页：闭包对应与内半连续

Semi-continuous Correspondences

Definition 3.1 Let X and Y be metric spaces, and let $\varphi : X \rightrightarrows Y$ be a correspondence.

(1) We say that φ is inner semi-continuous (i.s.c.) if whenever $x_n \to x$ and $y \in \varphi(x)$, there is a sequence $y_n \in \varphi(x_n)$ for each n such that $y_n \to y$.

(2) We say that φ is outer semi-continuous (o.s.c.) if whenever $x_n \to x$ and $y_n \in \varphi(x_n)$ for each n, there is a convergent subsequence $y_{n_k} \to y$ with $y \in \varphi(x)$.

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85 原课件第 85 页：闭包对应与内半连续

Semi-continuous Correspondences

Theorem 3.1 (Lower Semi-continuity and Inner Semi-continuity) Let X and Y be two metric spaces. A correspondence $\varphi : X \rightrightarrows Y$. φ is l.s.c. at x if and only if it is i.s.c. at x.

Proof. Suppose φ is not l.s.c. at some $x_0 \in X$. Then there exists an open subset G of Y such that $\varphi(x_0) \cap G \neq \varnothing$ and, for every $n \in \mathbb{N}$, there exists an $x_n \in B_{\frac{1}{n}}(x_0)$ with $\varphi(x_n) \cap G = \varnothing$. Note that $x_n \to x_0$, and pick any $y_0 \in \varphi(x_0) \cap G$. By hypothesis, there must exist a sequence $\{y_n : n \in \mathbb{N}\} \subseteq Y$ such that $y_n \to y_0$ and $y_n \in \varphi(x_n)$ for each n. But since $y_0 \in G$ and G is open, $y_n \in G$ for n large enough, contradicting that $\varphi(x_n) \cap G = \varnothing$ for all n.

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86 原课件第 86 页：闭包对应与内半连续

Semi-continuous Correspondences

Conversely, if $x_n \to x_0$ and $y_0 \in \varphi(x_0)$, then $y_0 \in \varphi(x_0) \cap B_{\frac{1}{n}}(y_0)$. By lower semi-continuity, there exists a $\varepsilon_n > 0$ with $\lim_{n\to\infty} \varepsilon_n = 0$ such that

$$x \in B_{\varepsilon_n}(x_0) \quad \Longrightarrow \quad \varphi(x) \cap B_{\frac{1}{n}}(y_0) \neq \varnothing.$$

Since $x_n \to x_0$, so there exists k_n such that for all $n \geqslant k_n$, $x_n \in B_{\varepsilon_n}(x_0)$, then $\varphi(x_n) \cap B_{\frac{1}{n}}(y_0) \neq \varnothing$. We may assume k_n is increasing.

For $n < k_n$, we pick $y_n \in \varphi(x_n)$. For any $k_n \leqslant n < k_{n+1}$, we have $\varphi(x_n) \cap B_{\frac{1}{n}}(y_0) \neq \varnothing$, pick $y_n \in \varphi(x_n) \cap B_{\frac{1}{n}}(y_0)$. Then $d(y_n, y_0) < \dfrac{1}{n}$. $y_n \to y_0$. Hence φ is i.s.c. at x_0. $\square$

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87 原课件第 87 页：外半连续的术语与序列选取

Semi-continuous Correspondences

Theorem 3.1′ (Upper Semi-continuity and outer Semi-continuity) Let X and Y be two metric spaces. Mapping $\varphi : X \rightrightarrows Y$ is a correspondence.

(1) If φ is compact-valued u.s.c. at x, then it is o.s.c. at x.

(2) If φ is o.s.c. at x, then φ is u.s.c. at x.

Proof of (1). Suppose φ is u.s.c. at x_0, $x_n \to x_0$ and $y_n \in \varphi(x_n)$. Since φ is compact-valued, $\varphi(x_0)$ has a bounded neighborhood U. Then $S = \{x_0, x_1, x_2, \cdots\}$ is sequentially compact in X, hence S is compact in X. By Proposition 1, $\varphi(S)$ is sequentially compact in Y. So, if $y_n \in \varphi(x_n)$ for each n, then $\{y_n : n \in \mathbb{N}\}$ — being a sequence in $\varphi(S)$ – must possess a subsequence that converges to some $y_0 \in \varphi(S)$. With an innocent abuse of notation, we denote this subsequence also by $\{y_n : n \in \mathbb{N}\}$, and write $y_n \to y_0$.

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本课此处的 o.s.c. 要求：任意 $x_n \to x$ 及 $y_n \in \Phi(x_n)$，有收敛子列且极限属于 $\Phi(x)$。它比文献中仅要求“已有收敛选择的极限不逃出”的闭图式 outer semicontinuity 更强。紧值上半连续给该性质：集合 $\{x, x_1, x_2, \ldots\}$ 紧，紧像定理给子列，闭图给极限归属。反向若上半连续失效，选开 $G \supset \Phi(x)$ 及逃出 G 的 y_n，由假设抽子列极限在 $\Phi(x)$，与开性矛盾。复合用先抽中间值再抽末端值，有限积用有限次子列抽取；不能无限次抽取后还把未构造的子列当作存在。

88 原课件第 88 页：外半连续的术语与序列选取

Semi-continuous Correspondences

We wish to show that $y_0 \in \varphi(x_0)$. Let $y_0 \notin \varphi(x_0)$, and observe that this implies $\varepsilon = d_Y(y_0, \varphi(x_0)) > 0$ because $\varphi(x_0)$ is a compact (hence closed) set. Now define

$$T = \left\{ z \in Y : d_Y(z, \varphi(x_0)) \leqslant \frac{\varepsilon}{2} \right\}$$

Since $d_Y(z, \varphi(x_0))$ is a continuous map on z, T is a closed set. Moreover, by definition, we have

$$\varphi(x_0) \subseteq T^\circ$$

and $y_0 \notin T$. But, since φ is u.s.c. at x_0, there exists $\delta > 0$ such that

$$\varphi(B_\delta(x_0)) \subseteq T^\circ.$$

Thus there is $n \in \mathbb{N}$ such that $y_n \in \varphi(x_n) \subseteq T^\circ$ for all $n \geqslant N$. But then, since T is a closed set and $y_n \to y_0$, we must have $y_0 \in T$, a contradiction.

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89 原课件第 89 页：外半连续的术语与序列选取

Semi-continuous Correspondences

We wish to show that $y_0 \in \varphi(x_0)$. Let $y_0 \notin \varphi(x_0)$. Note that $\varphi(x_0)$ is compact, there exist two neighbourhoods $V(y_0)$ and $V'(\varphi(x_0))$ such that $V(y_0) \cap V'(\varphi(x_0)) = \varnothing$, then $y_0 \notin \overline{V'}(\varphi(x_0))$.

If φ is u.s.c. at x_0 and $x_n \to x_0$, then there exists a $N \in \mathbb{N}$ such that $\varphi(x_n) \subseteq V'(\varphi(x_0))$ for $n \geqslant N$, thus $y_n \in \overline{V'}(\varphi(x_0))$ for $n \geqslant N$, and $y_0 \in \overline{V'}(\varphi(x_0))$ for $n \geqslant N$, which contradicts $y_0 \notin \overline{V'}(\varphi(x_0))$.

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90 原课件第 90 页：外半连续的术语与序列选取

Semi-continuous Correspondences

Proof of (2). Now suppose that for every sequence $x_n \to x_0$ and $y_n \in \varphi(x_n)$, there is a subsequence of y_n with limit in $\varphi(x)$.

Suppose φ is not u.s.c., then there exists an open subset G of Y with $\varphi(x_0) \subseteq G$ and

$$\varphi\left(B_{\frac{1}{n}}(x_0)\right) \setminus G \neq \varnothing$$

for any $n \in \mathbb{N}$. But then there exist $(x_n) \subseteq X$ and $(y_n) \subseteq Y$ such that $x_n \to x_0$ and $y_n \in \varphi\left(B_{\frac{1}{n}}(x_0)\right) \setminus G$ for each n. Since $\varphi(x_0) \subseteq G$ and each y_n belongs to the closed set $Y \setminus G$, no subsequence of (y_n) can converge to a point in $\varphi(x_0)$. □

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91 原课件第 91 页：外半连续的术语与序列选取

Semi-continuous Correspondences

Proof of (2). Now suppose that for every sequence $x_n \to x_0$ and $y_n \in \varphi(x_n)$, there is a subsequence of y_n with limit in $\varphi(x)$.

Suppose φ is not u.s.c., then for any open set $G \subseteq Y$ with $\varphi(x_0) \subseteq G$ and any neighbourhood $U_n(x_0)$, there exist $x_n \in U_n(x_0)$ such that $\varphi(x_n) \nsubseteq G$. As $x_n \to x_0$, there exists $y_n \in \varphi(x_n)$ and $y_n \notin G$. Then there exists a convergent subsequence $\{y_{n'}; n' \in \mathbb{N}\} \subseteq \{y_n; n \in \mathbb{N}\}$ such that $y_{n'} \to \varphi(x_0)$. However, $y_n \notin G$, there exists a convergent subsequence $\{y_{n'}; n' \in \mathbb{N}\} \subseteq \{y_n; n \in \mathbb{N}\}$ such that $\lim_{n' \to \infty} y_{n'} \notin G$, then $\lim_{n' \to \infty} y_{n'} \notin \varphi(x_0)$. This is a contradiction! □

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92 原课件第 92 页：外半连续的术语与序列选取

Properties of Semi-continuous Mappings

Theorem 3.2. (Inner Semi-continuity) If φ_1 is an i.s.c. mapping of X into Y and φ_2 is an i.s.c. mapping of Y into Z, then the composition product $\varphi = \varphi_2 \cdot \varphi_1$ is an i.s.c. mapping of X into Z.

Proof. If $x_n \to x_0$ and $z_0 \in \varphi(x_0) = (\varphi_2 \cdot \varphi_1)(x_0) = \varphi_2(\varphi_1(x_0))$, then there exists $y_0 \in \varphi_1(x_0)$ such that $z_0 \in \varphi_2(y_0)$.

Whenever $x_n \to x_0$ and $y_0 \in \varphi_1(x_0)$, by inner semi-continuity of φ_1, there exists a sequence $y_n \in \varphi_1(x_n)$ such that $y_n \to y_0$. On the other hand, $z_0 \in \varphi_2 y_0$, by inner semi-continuity of φ_2, there exists a sequence $z_n \in \varphi_2(y_n)$ such that $z_n \to z_0$.

$[y_n \in \varphi_1(x_n)$ and $z_n \in \varphi_2(y_n)]$ imply $z_n \in \varphi_2(\varphi_1(x_n)) = \varphi(x_n)$, thus $\varphi = \varphi_2 \cdot \varphi_1$ is i.s.c. $\square$

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93 原课件第 93 页：外半连续的术语与序列选取

Properties of Semi-continuous Mappings

Theorem 3.2′ (Outer Semi-continuity) If φ_1 is an o.s.c. mapping of X into Y and φ_2 is an o.s.c. mapping of Y into Z, the composition product $\varphi = \varphi_2 \cdot \varphi_1$ is an o.s.c. mapping of X into Z.

Proof. If $x_n \to x_0$ and $z_n \in \varphi(x_n) = (\varphi_2 \cdot \varphi_1)(x_n) = \varphi_2(\varphi_1(x_n))$, then there exists $y_n \in \varphi_1(x_n)$ such that $z_n \in \varphi_2(y_n)$.

Whenever $x_n \to x_0$ and $y_n \in \varphi_1(x_n)$, by outer semi-continuity of φ_1, there exists a convergent subsequence $y_{n_1} \to y_0$ with $y_0 \in \varphi_1(x_0)$. On the other hand, $z_n \in \varphi_2(y_n)$, then $z_{n_1} \in \varphi_2(y_{n_1})$. By outer semi-continuity of φ_2, there exists a convergent subsequence $z_{n_2} \to z_0$ with $z_0 \in \varphi_2(y_0)$.

$[y_0 \in \varphi_1(x_0)$ and $z_0 \in \varphi_2(y_0)]$ imply $z_0 \in \varphi_2(\varphi_1(x_0)) = \varphi(x_0)$, thus $\varphi = \varphi_2 \cdot \varphi_1$ is o.s.c. □

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94 原课件第 94 页：外半连续的术语与序列选取

Semi-continuous Correspondences

Theorem 3.3 (Inner Semi-continuity) The Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ of a finite number of i.s.c. mappings φ_n of X into Y_n is an i.s.c. mapping of X into $Y = \prod_{n=1}^{N} Y_n$.

Proof. If $x^q \to x^0$ and $\left(y_1^0, \cdots, y_N^0\right) = y^0 \in \varphi\left(x^0\right) = \prod_{n=1}^{N} \varphi_n\left(x^0\right)$, then $y_n^0 \in \varphi_n\left(x^0\right)$ for $n = 1, \cdots, N$. If every φ_n is i.s.c., then there is a sequence $y_n^q \in \varphi_n\left(x^q\right)$ for each q such that $y_n^q \to y_n^0$. Thus $y^q = \left(y_1^q, \cdots, y_N^q\right) \in \varphi\left(x^q\right) = \prod_{n=1}^{N} \varphi_n\left(x^q\right)$ such that $y^q = \left(y_1^q, \cdots, y_N^q\right) \to \left(y_1^0, \cdots, y_N^0\right) = y^0$. Therefore $\varphi = \prod_{n=1}^{N} \varphi_n$ is i.s.c. □

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95 原课件第 95 页：外半连续的术语与序列选取

Semi-continuous Correspondences

Theorem 3.3′ **(Outer Semi-continuity)** The Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ of a finite number of o.s.c. mappings φ_n of X into Y_n is an o.s.c. mapping of X into $Y = \prod_{n=1}^{N} Y_n$.

Proof. If $x^q \to x^0$ and $(y_1^q, \cdots, y_N^q) = y^q \in \varphi(x^q) = \prod_{n=1}^{N} \varphi_n(x^q)$, then $y_n^q \in \varphi_n(x^q)$ for $n = 1, \cdots, N$. If φ_1 is o.s.c., then there is a convergent subsequence $y_1^{q_1} \to y_1^0$ with $y_1^0 \in \varphi_1 x^0$. In this case, $y_n^{q_1} \in \varphi_n x^{q_1}$ for $n = 1, \cdots, N$. If φ_2 is o.s.c., then there is a convergent subsequence $y_n^{q_2} \to y_n^0$ with $y_n^0 \in \varphi_n(x^0)$ for $n = 1, 2$. In this case, $y_n^{q_2} \in \varphi_n x^{q_2}$ for $n = 1, \cdots, N$. Follow these steps, we can obtain that, if φ_N is o.s.c., then there is a convergent subsequence $y_n^{q_N} \to y_n^0$ with $y_n^0 \in \varphi_n(x^0)$ for $n = 1, \cdots, N$. Hence $y^{q_N} = (y_1^{q_N}, \cdots, y_N^{q_N}) \to (y_1^0, \cdots, y_N^0) = y^0$ and $y^0 = (y_1^0, \cdots, y_N^0) \in \prod_{n=1}^{N} \varphi_n(x^0) = \varphi(x^0)$. Therefore $\varphi = \prod_{n=1}^{N} \varphi_n$ is o.s.c. □

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96 原课件第 96 页：最优值的两种半连续性

Berge's Maximum Theorem

Theorem 4.1. If f is a l.s.c. numerical function in $S \times T$ and φ is a l.s.c. mapping of S into T such that, for each x, $\varphi(x) \neq \varnothing$, the numerical function M defined by

$$M(x) = \sup\{f(x,y) \mid y \in \varphi(x)\}$$

is l.s.c.

Proof. Suppose that $x_0 \in S$ and let y_0 be such that

$$y_0 \in \varphi(x_0); \qquad f(x_0, y_0) \geqq M(x_0) - \varepsilon.$$

There exist neighbourhoods $U(x_0)$ and $V(y_0)$ such that

$$(x,y) \in U(x_0) \times V(y_0) \Longrightarrow f(x,y) \geqq f(x_0,y_0) - \varepsilon \geqq M(x_0) - 2\varepsilon;$$

there exists a neighbourhood $U'(x_0)$ such that

$$x \in U'(x_0) \Longrightarrow \varphi(x) \cap V(y_0) \neq \varnothing.$$

Therefore

$$x \in U(x_0) \cap U'(x_0) \Longrightarrow M(x) \geqq M(x_0) - 2\varepsilon.$$

□

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设 $M(x) = \sup_{y \in \Phi(x)} f(x,y)$。若 f 下半连续且 Φ 下半连续，任取 $a < M(x)$，选可行 y 满足 $f(x,y) > a$。目标下半连续给乘积邻域使函数值仍大于 a，可行对应下半连续给附近可行点落入该邻域，故附近 $M > a$。若 f 上半连续，Φ 非空紧值上半连续，则每点最大值存在。取输入序列及最优点，由紧像抽子列趋于可行极限，上半连续给 $\limsup M(x_n) \leq M(x)$。对 limsup 严格写法需先选逼近它的子列，再抽最优点子列。

97 原课件第 97 页：最优值的两种半连续性

Berge's Maximum Theorem

Theorem 4.2. If f is a u.s.c. numerical function in $S \times T$ and φ is a compact-valued u.s.c. correspondence of S into T such that, for each x, $\varphi(x) \neq \varnothing$, the mumerical function M defined by

$$M(x) = \max\{f(x, y) \mid y \in \varphi(x)\}$$

is u.s.c.

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本页继续“最优值的两种半连续性”。完整定义、条件和论证见本讲原页第 96 页下方。

98 原课件第 98 页：最优值的两种半连续性

Berge's Maximum Theorem

Proof. Suppose that $x_0 \in S$; to each y in $\varphi(x_0)$ there correspond neighbourhoods $U_y(x_0)$ and $V(y)$ such that

$$(x,z) \in U_y(x_0) \times V(y) \quad \Longrightarrow \quad f(x,z) \leqq f(x_0,y) + \varepsilon.$$

Since $\varphi(x_0)$ is compact, it can be covered by a finite number of neighbourhoods of the form $V(y)$, say $V(y_1), V(y_2), \cdots, V(y_N)$.

Putting $U'(x_0) = \bigcap_{n=1}^{N} U_{y_n}(x_0)$ and $V(\varphi(x_0)) = \bigcup_{n=1}^{N} V(y_n)$, we have

$$x \in U'(x_0), y \in V(\varphi(x_0)) \Longrightarrow f(x,y) \leqq \max_n f(x_0,y_n) + \varepsilon \leqq M(x_0) + \varepsilon.$$

Moreover there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \subseteq V(\varphi(x_0))$$

and so

$$x \in U(x_0) \cap U'(x_0) \quad \Longrightarrow \quad M(x) = \max_{y \in \varphi(x)} f(x,y) \leqq M(x_0) + \varepsilon.$$

□

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“最优值的两种半连续性”。完整定义、条件和论证见本讲原页第 96 页下方。

99 原课件第 99 页：Berge 最大值与唯一选择

Berge's Maximum Theorem

Theorem 4.3 (Maximum Theorem). If f is a continuous numerical function in T and φ is a continuous mapping of S into T such that, for each x, $\varphi(x) \neq \varnothing$, then the numerical function M defined by $M(x) = \max\{f(y) \mid y \in \varphi(x)\}$ is continuous in X and the mapping f defined by $f(x) = \{y \in \varphi(x) \mid f(y) = M(x)\}$ is a u.s.c. correspondence of S into T.

Proof. The function f is continuous in $S \times T$ and so M is a continuous numerical function; moreover the mapping Δ given by

$$\Delta x = \{y \in T \mid M(x) - f(y) \leqq 0\}$$

is closed and hence $f = \varphi \cap \Delta$ is u.s.c. $\square$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

必须补充非空紧值：Φ 连续且非空紧值，f 连续，则第96页的两个方向给 M 连续。最优对应 $A(x) = \{y \in \Phi(x) : f(x,y) = M(x)\}$ 非空紧；等式由连续性保极限，故闭图，再以 Φ 为紧容器得上半连续。若最优点唯一，单点值上半连续等价普通连续，故唯一最优选择连续。实数紧值取 $f(y) = y$，最大值唯一，得到连续选择；没有紧值时例如 $(0,1)$ 根本没有最大点。原页100的球应理解为非空闭球，开球一般不取到坐标最大值。

100 原课件第 100 页：Berge 最大值与唯一选择

Berge's Maximum Theorem

Let f be a continuous numerical function defined in a topological space T. A family of compact sets $\mathscr{K} = \{K_i \mid i \in I\}$ in T is called selective (with respect to f) if for each i there exists one and only one a_i such that

$$a_i \in K_i; \qquad f(a_i) = \max\{f(y) \mid y \in K_i\}.$$

In other words, the maximum of f is attained at only one point of the set K_i. For example, in $\mathbb{R}^N$ every family of balls is selective, with respect to $f(y) = \pi_i y$; in $\mathbb{R}$, every family of compact sets is selective with respect to $f(y) = y$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“Berge 最大值与唯一选择”。完整定义、条件和论证见本讲原页第 99 页下方。

101 原课件第 101 页：Berge 最大值与唯一选择

Berge's Maximum Theorem

Theorem 4.4. Let φ be a continuous mapping of S into T such that, for each x, $\varphi(x) \neq \varnothing$. If the family of sets $\{\varphi(x) \mid x \in S\}$ is selective, there is a single-valued continuous mapping σ of S into T such that, for each x, $\sigma(x) \in \varphi(x)$.

Proof. Let f be a continuous numerical function in T for which $\{\varphi(x) \mid x \in S\}$ is selective; if $\phi(x) = \{y \in \varphi(x) \mid f(y) = M(x)\}$ then ϕ is a single-valued mapping of S into T. Moreover, it is u.s.c. by the preceding theorem and so is continuous (since it is single-valued). The mapping $\sigma(x) = \phi(x)$ satisfies

$$\sigma(x) \in \varphi(x) \quad \text{for all} \quad x \in S$$

as required. $\square$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“Berge 最大值与唯一选择”。完整定义、条件和论证见本讲原页第 99 页下方。

102 原课件第 102 页：Berge 最大值与唯一选择

Berge's Maximum Theorem

Corollary 4.1. If φ is a continuous mapping of X into $\mathbb{R}$ such that, for each x, $\varphi(x) \neq \varnothing$, there exists a continuous single-valued mapping σ such that, for each x, $\sigma(x) \in \varphi(x)$

Proof. It is sufficient to take $f(y) = y$, whence $\sigma(x) = \max \varphi(x)$. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“Berge 最大值与唯一选择”。完整定义、条件和论证见本讲原页第 99 页下方。

103 原课件第 103 页：最小化的对偶证明

Berge's Maximum Theorem

Theorem 4.5. If f is a l.s.c. numerical function in $S \times T$ and φ is a compact-valued u.s.c. correspondence of S into T such that, for each x, $\varphi(x) \neq \varnothing$, the numerical function M defined by

$$M(x) = \inf\{f(x, y) \mid y \in \varphi(x)\}$$

is l.s.c.

Proof. We want to prove that, for any $\varepsilon > 0$, there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \Longrightarrow M(x_0) - \varepsilon \leqslant M(x).$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

对最小化使用 $-f$ 的最大化，或直接证明：紧值上半连续可行集及下半连续目标给最小值函数下半连续；下半连续可行集及上半连续目标给下确界函数上半连续。后一证明先选 y_0 使 $f(x_0, y_0) < m(x_0) + \varepsilon/2$，再使附近某可行选择的目标小于 $f(x_0, y_0) + \varepsilon/2$。两项合成 $m(x) < m(x_0) + \varepsilon$。第104页末尾应是 $M(x)$，不是 $M(x_0)$；第106页错误的集合包含应改成存在 $y \in \Phi(x) \cap V$。第107页最小对应的等式最安全写为 $f(y) = m(x)$，原图弱不等式方向不能把所有较大值也选入。

104 原课件第 104 页：最小化的对偶证明

Berge's Maximum Theorem

If φ is compact-valued, then there exist $y_1, \cdots, y_K \in \varphi(x_0)$ such that $\varphi(x_0) \subseteq \bigcup_{k=1}^{K} V_k(y_k)$. If φ is u.s.c., then there exists a neighbourhood $U_0(x_0)$ such that

$$x \in U_0(x_0) \Longrightarrow \varphi(x) \subseteq \bigcup_{k=1}^{K} V_k(y_k).$$

If f is l.s.c., then, for any $\varepsilon > 0$, $y_k \in \varphi(x_0)$, there exists a neighbourhood $U_{y_k}(x_0)$ such that, for any $x \in U_{y_k}(x_0)$ and any $y \in V_k(y_k)$, $f(x_0, y_k) - \varepsilon \leqslant f(x, y)$.

Define $U(x_0) = U_0(x_0) \cap \bigcap_{k=1}^{K} U_{y_k}(x_0)$. Note that $y_1, \cdots, y_K \in \varphi(x_0)$, then, for any $x \in U(x_0)$ and $y \in \varphi(x)$,
$M(x_0) - \varepsilon \leqslant \inf_{y_k \in \varphi(x_0)} f(x_0, y_k) - \varepsilon \leqslant f(x, y)$.

Therefore, for any $x \in U(x_0)$ and $y \in \varphi(x)$,
$M(x_0) - \varepsilon \leqslant \inf_{y \in \varphi(x)} f(x, y) = M(x_0)$. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“最小化的对偶证明”。完整定义、条件和论证见本讲原页第 103 页下方。

105 原课件第 105 页：最小化的对偶证明

Berge's Maximum Theorem

Theorem 4.6. If f is a u.s.c. numerical function in $S \times T$ and φ is a l.s.c. mapping of S into T such that, for each x, $\varphi(x) \neq \varnothing$, the numerical function M defined by

$$M(x) = \inf\{f(x, y) \mid y \in \varphi(x)\}$$

is u.s.c.

Proof. We want to prove that, for any $\varepsilon > 0$, there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \Longrightarrow M(x) \leqslant M(x_0) + \varepsilon.$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“最小化的对偶证明”。完整定义、条件和论证见本讲原页第 103 页下方。

106 原课件第 106 页：最小化的对偶证明

Berge's Maximum Theorem

From the definition of infimum, for any $\varepsilon > 0$, there exists $y_0 \in \varphi(x_0)$ such that $f(x_0, y_0) \leqslant M(x_0) + \frac{\varepsilon}{2}$.
If f is u.s.c., then there exist two neighbourhoods $U_1(x_0)$ and $V(y_0)$ such that, for any $x \in U_1(x_0)$ and any $y \in V(y_0)$, $f(x, y) \leqslant f(x_0, y_0) + \frac{\varepsilon}{2}$.
Therefore $f(x, y) \leqslant f(x_0, y_0) + \frac{\varepsilon}{2} \leqslant M(x_0) + \varepsilon$.
If φ is l.s.c., for $\varphi(x_0) \cap V(y_0) \neq \varnothing$, then there exists a neighbourhood $U_2(x_0)$ such that

$$x \in U_2(x_0) \Longrightarrow \varphi(x) \subseteq \varphi(x) \cap V(y_0) \neq \varnothing.$$

Define $U(x_0) = U_1(x_0) \cap U_2(x_0)$. for any $x \in U(x_0)$, there exists $y \in V(y_0)$, such that

$$M(x) = \inf_{y \in \varphi(x)} f(x, y) \leqslant f(x, y) \leqslant f(x_0, y_0) + \frac{\varepsilon}{2} \leqslant M(x_0) + \varepsilon.$$

Therefore M is u.s.c. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“最小化的对偶证明”。完整定义、条件和论证见本讲原页第 103 页下方。

107 原课件第 107 页：最小化的对偶证明

Berge's Maximum Theorem

Theorem 4.7 (Maximum Theorem). If f is a continuous numerical function in T and φ is a continuous mapping of S into T such that, for each x, $\varphi(x) \neq \varnothing$, then the numerical function M defined by $M(x) = \min\{f(y) \mid y \in \varphi(x)\}$ is continuous in X and the mapping f defined by $\phi(x) = \{y \in \varphi(x) \mid f(y) = M(x)\}$ is a u.s.c. correspondence of S into T.

Proof. The function f is continuous in $S \times T$ and so M is a continuous numerical function; moreover the mapping Δ given by

$$\Delta x = \{y \in T \mid M(x) - f(y) \leqq 0\}$$

is closed and hence $\phi = \varphi \cap \Delta$ is u.s.c. $\square$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“最小化的对偶证明”。完整定义、条件和论证见本讲原页第 103 页下方。

108 原课件第 108 页：环境依赖偏好与最大元

Berge's Maximum Theorem

In economic activities, decision makers often change their preferences according to the situation of the environment, so the preference in economic activities is often changed by the environment on which preferences depend. Suppose $\beta : S \to T$ is a set-valued mapping, for any $x \in S$, $\prec_x$ is a binary relationship defined on $\beta(x)$ (set-valued mapping), which is transitive, but is not reflexive and complete. Set $\varphi(x)$ to be the set of maximal elements in $\beta(x)$ according to $\prec_x$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

假设 β 非空紧值连续，每个严格关系传递且非自反，且“可行且非严格优于”的三元组集合闭。若无极大元，每个 y 被某个 z 严格压过，这些严格下截集构成 $\beta(x)$ 的相对开覆盖。有限子覆盖的有限个 z_i 中存在极大者（传递且无循环），它不被任一 z_i 压过，矛盾。极大元集为所有非优于截集的交，故闭且紧。若 $x_n \to x, y_n$ 为极大元且 $y_n \to y$，先由可行图闭得 $y \in \beta(x)$，再对任意 $z \in \beta(x)$ 用下半连续选 $z_n \in \beta(x_n) \to z$。由 y_n 不差于 z_n 及三元组闭性得极限不被 z 压过。第111页 z_n 应属于可行集 $\beta(x_n)$，不必属于最优集。以连续效用定义严格偏好即可回到 Berge 定理；第112页最优对应内目标须写成 $f(x, y)$，并补紧值条件。

109 原课件第 109 页：环境依赖偏好与最大元

Berge's Maximum Theorem

Theorem 4.8. Suppose that $\beta : S \to T$ is a correspondence, $\varphi(x) = \left\{ y \in T \mid y \in \max_{\prec_x} \beta(x) \right\}$, and $\{(x, y, z) \in S \times T \times T \mid y \in \beta(x), z \in \beta(x), y \not\prec_x z\}$ is closed. If $\beta(x)$ is compact, then $\varphi(x) \neq \varnothing$ is compact, β is continuous, φ is u.s.c.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“环境依赖偏好与最大元”。完整定义、条件和论证见本讲原页第 108 页下方。

110 原课件第 110 页：环境依赖偏好与最大元

Berge's Maximum Theorem

Proof. If $\varphi(x) = \varnothing$, then for any $y \in \beta(x)$, there exists $y' \in \beta(x)$ such that $y \prec_x y'$. Define $r(z) = \{y \in \beta(x) \mid y \prec_x z\}$, then $\beta(x) \subseteq \bigcup_{z \in \beta(z)} \{r(z)\}$. Note that $\beta(x)$ is compact and $\bigcup_{z \in \beta(z)} \{r(z)\}$ is its open covering, then there exist $r(z_1), \cdots, r(z_K)$, such that $\beta(x) \subseteq \bigcup_{k=1}^{K} r(z_k)$. $\{z_1, \cdots, z_K\}$ in finite, then there is the maximal to be z_K, as $\prec_x$ is transitive, then for $k = 1, \cdots, K-1$, $z_K \notin r(z_k)$. But $\prec_x$ is not reflexive, then $z_K \notin r(z_K)$, and hence $z_K \notin \beta(x)$, which is a contradiction. Therefore $\varphi(x) \neq \varnothing$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“环境依赖偏好与最大元”。完整定义、条件和论证见本讲原页第 108 页下方。

111 原课件第 111 页：环境依赖偏好与最大元

Berge's Maximum Theorem

According to the continuity of $\prec_x$, $\{y \in \beta(x) \mid y \not\prec_x z\}$ is closed, then

$$\varphi(x) = \{y \in T \mid z \in \beta(x), y \not\prec_x z\} = \bigcap_{z \in \beta(x)} \{y \in \beta(x) \mid y \not\prec_x z\}$$

is closed, $\varphi(x) \neq \varnothing$ and $\varphi(x) \subseteq \beta(x)$, $\beta(x)$ is compact, then $\varphi(x)$ is compact.

In order to check that φ is u.s.c.at x_0, we want to prove that φ is closed-valued,

$$\text{for any } (x_n, y_n) \to (x, y),\ y_n \in \varphi(x_n) \quad \Longrightarrow \quad y \in \varphi(x).$$

Since β is l.s.c., then, for $z \in \beta(x)$, there exists $z_n \to z$ such that, for any n, $z_n \in \varphi(x_n)$, then $y_n \not\prec_{x_n} z_n$. From the continuity of $\prec_x$, $x_n \to x$, $y_n \to y$, $z_n \to z$, then $y \not\prec_x z$, and hence $y = \max_{\prec_x} \beta(x)$, that is, $y \in \varphi(x)$, thus φ is u.s.c. $\square$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“环境依赖偏好与最大元”。完整定义、条件和论证见本讲原页第 108 页下方。

112 原课件第 112 页：环境依赖偏好与最大元

Berge's Maximum Theorem

Corollary 4.2. If f is a continuous numerical function in $S \times T$ and φ is a continuous mapping of S into T such that, for each x, $\varphi(x) \neq \varnothing$, the numerical function M defined by

$$M(x) = \max\{f(x, y) \mid y \in \varphi(x)\}$$

is continuous in X and the mapping ϕ defined by

$$\phi(x) = \{y \in \varphi(x) \mid f(y) = M(x)\}$$

is u.s.c. of S into T.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“环境依赖偏好与最大元”。完整定义、条件和论证见本讲原页第 108 页下方。

113 原课件第 113 页：环境依赖偏好与最大元

Berge's Maximum Theorem

Proof. Define

$$y \prec_x z \qquad \Longleftrightarrow \qquad f(x,y) \prec f(x,z)$$

Since f is continuous, then
$\{(x,y,z) \in S \times T \times T \mid f(x,y) \not\prec f(x,z)\}$ is closed, then for any $x \in S$ and $y \in Y$, $\{z \in Y \mid f(x,y) \prec_x f(x,z)\}$ is closed, then f is u.s.c.

We now check that M is continuous. $\{x\}$ is a single-point mapping of correspondence, the product mapping $(x, r(x))$ is u.s.c., f is continuous, then $f(x, r(x))$ is u.s.c., but $f(x, r(x)) = \{M(x)\}$, then $\alpha : S \to \mathbb{R}$ is continuous. $\square$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“环境依赖偏好与最大元”。完整定义、条件和论证见本讲原页第 108 页下方。

补充：凸包所用的 Carathéodory 定理

在 $\mathbb{R}^N$，每个凸包点都能用至多 $N+1$ 个原集合点表示。证明：从任意有限表示 $x=\sum_{i=1}^{m}\lambda_i x_i$ 开始，去掉零权重。若 $m>N+1$，向量 $(x_i,1)\in\mathbb{R}^{N+1}$ 线性相关，故有不全零 a_i 满足 $\sum_i a_i x_i=0,\sum_i a_i=0$。正负 a_i 均存在，取 $t=\min_{a_i>0}\lambda_i/a_i>0$，新权重 λ_i-ta_i 均非负、和仍1、表示的点不变，且至少一个权重变0。去掉该点，有限次后至多剩 $N+1$ 点。因此紧集凸包是紧单纯形乘以 K^{N+1} 在连续凸组合映射下的像，仍紧。

证明自测 为什么最大值定理只给最优对应上半连续？例目标 $f(x,y)=xy$，可行集恒为 $[-1,1]$。$x=0$ 时全部点最优，$x>0$ 时仅1最优，$x<0$ 时仅 -1 最优；原点的旧最优点0无法由正参数最优点逼近，所以下半连续失败。最优值为 $|x|$，仍连续。